
The Formal-Alone Theory of Everything

A Constitutive Theory of Objects and Observations under Registration, Price, Crossing, and Cost

Research Paper

Constitutive register: map the structure, price the boundaries, file the unclosed.

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Abstract. An open problem is priced here as a measurement, not mourned as a mystery. On a finite frame of data, an involution, a reading class, and a Boolean target, three theorems are proved in two lines apiece and machine-checked in core Lean 4: no even reading decides an odd target (the Wall); in the full wholly-odd product family the missing content is exactly one binary coordinate per free orbit, and for a constrained row the exact price of an exhibited family A is $\log_2|A|$ in information, $\lceil \log_2|A| \rceil$ in fixed-length bits, one exactly when $|A| = 2$ (the Price); and one odd supply beside one even calibration decides the target, with uniqueness (the Crossing). The torsor of odd sections over even calibrations is proved, and the sign-blindness of the register's kernel is executed, not argued: the Gram map is an invariant observation of its frame under the flip — not a projection; it is nonlinear, non-idempotent, and its fibres run larger than any single reflection pair — bit-identical under reflection while the direction flips, and therefore unable to recover determinant sign. One conjecture extends the theorems to every blocked row: the One-Cut Hypothesis, that resolution enters as one provenance-clean orientation supply, dated at entry. It is armed with six falsifiers and two predictions, including a height-pairing computation at analytic order two on the Birch and Swinnerton-Dyer row, smallest test curve 389.a1. A dated-entry recursion on the lattice of corpora admits only finitely many strict observation refinements on a finite frame — a stabilization bound, not a promise of arrival at the discrete partition — and its termination typing recovers the species dichotomy of the census; the port of entry is a single diagonal read as wall and executed as self. Twenty-three rows across complexity, number theory, analysis, geometry, and topology are typed under one anatomy, with ten dated crossings as non-vacuity controls and the Perelman crossing as the grand witness. A kinetic demonstration prices registration at the Landauer floor. No open problem is solved; the shape of every solution is named — at hypothesis grade, as the One-Cut Hypothesis, not as a proved theorem about corpora.

Keywords *formal-alone theory, odd-supply separation, one-cut hypothesis, register blindness, orientation supply, provenance, dated-entry recursion, Ninth Gate, sign-blind kernel, Landauer floor, Millennium Prize problems, Birch and Swinnerton-Dyer conjecture*

An open problem is a blocked register, priced at one missing bit per unresolved orbit

The sentence above is the One-Cut Hypothesis: the theory's single conjecture, armed with six falsifiers (§8). The theorems price every certified row (X, τ, ρ, A) at $\log_2|A|$, one bit exactly when $|A| = 2$.

1. Introduction: the claim, whole

Mathematics keeps its hard problems in twenty different vocabularies. Complexity theory speaks of oracle barriers, sieve theory of the parity problem, proof theory of independence, analysis of supercriticality. This treatise replaces all of them with one measurement.

Take any problem and fix the register that works on it: the methods, the readings, the proof techniques, idealized and closed under their own moves. Every such register carries a symmetry, the flip it cannot see: oracle attachment, sign reversal, parity exchange, orientation reversal. Call a reading **even** when the flip leaves it unchanged and a target **odd** when the flip changes it. Then:

1. **The Wall** (theorem). No even reading, and no finite combination of even readings, decides an odd target. Ever. For any register, in any world with finite data and a symmetry.

2. **The Price** (theorem). In the full wholly-odd product family, the deficit is exactly one binary orientation coordinate per free orbit — the fibre has exactly 2^m members on m orbits. For a constrained row, the exact deficit is $\log_2|A|$, with a one-bit crossing exactly when $|A| = 2$. Never a magnitude, never a continuum.
3. **The Crossing** (theorem). At a one-bit seat, fix an aligned wholly odd reference section, entering from outside the register. The unique orbit-even calibration selecting one of the two admissible orientations is the supplied bit; the target is then reconstructed on the orbit. Nothing else does. Nothing else is needed.

These three theorems are proved in Sections 4–6 in two lines apiece, and they are machine-checked in Lean 4 (Appendix A). Together they imply the theory’s single sentence:

An open problem is not a hard problem. It is a blocked register missing one bit per unresolved orbit, and the whole craft of resolution is the arrival of that bit from outside — seat by seat.

The epigraph states the conjecture at conjecture grade; what this treatise proves is the anatomy underneath it — wall, price, and crossing-shape — on every exhibited frame.

One conjecture extends the theorems to every problem row (Section 8). One postulate grounds the physical interface (Section 13). Both are stated at face value. The remaining load split is declared, not blurred: the finite spine T1–T15 is machine-checked in core Lean (Appendix A.1), and so are the general Price bijection T16 and the Boolean oddness equivalence T17 (Appendix A.2), while the finite cardinal corollary of T16 is executed at the K4 frame and not derived; the census tables are structural diagnostics; Appendix C is documentary. No sentence in this treatise grades itself above that split.

The theory is three bits, read in order and never compressed into one. The first bit is theory: on a finite frame (X, τ, d, C) with $\tau^2 = \text{id}$, if every reading in C is even under τ and the target d is odd, then no finite coalition of readings decides d . The proof is two lines: evenness gives $r(\tau x) = r(x)$; oddness gives $d(\tau x) \neq d(x)$; closure under pairing preserves evenness. The second bit is hypothesis: the One-Cut Hypothesis, that for a certified blocked row there exists one provenance-clean orientation supply, odd under the row’s flip, not denotable by the dated register on each orbit, whose dated entry closes the missing coordinate. It is named at hypothesis grade and armed with falsifiers. The third bit is witness: resolution itself has a theory. A corpus is dated; a supply enters by deed; each provenance-clean independent supply that is nonconstant on some current fiber block strictly refines the reading class’s fiber partition, so on a finite frame the number of strict refinements is bounded by $|U|$ minus the initial block count and every such chain halts. Halting is stabilization, not arrival: the terminal partition need not be discrete, and a target varying inside a terminal block remains unresolved. Where complete resolution is intended, it carries an explicit availability hypothesis — whenever the current observation does not determine the target, an admissible supply exists that strictly refines a block on which the target varies — and that hypothesis is exactly the One-Cut content per row, never a theorem of finiteness. Crossed halt is common species. Defeater-absent halt is diagonal species. The port is the Ninth Gate: one diagonal, two surfaces, read as wall and executed as self.

The claim is exact: an open row is not addressed by force inside a proved-inert register. It is addressed by a dated supply the register could not denote, entering once, at one seat, with one bit of orientation. The theory does not solve a Millennium row. It prices the shape of resolution and disciplines the bought premise.

2. The setup, complete

Let X be a finite set of data, $\tau: X \rightarrow X$ an involution ($\tau^2 = \text{identity}$), the flip the register cannot see. A **reading** is a function $f: X \rightarrow \{0,1\}$; it is **even** when $f(\tau x) = f(x)$ for all x , meaning it factors through the orbit space X/τ . A **target** is a function $d: X \rightarrow \{0,1\}$; it is **odd at x** when $d(\tau x) \neq d(x)$, and **wholly odd** when $d(\tau x) = \neg d(x)$ for all x . Two strengths, named and never interchanged. For Boolean targets, wholly odd is equivalent to odd at every point, because two distinct Boolean values are complements (T13_wholly_odd_is_odd_everywhere machine-checks the forward direction; the full iff is T17, machine-checked in Appendix A.2 by a one-line case analysis). What is false, and claimed nowhere, is the step from oddness at one designated point to whole-domain anti-invariance: the Wall needs only oddness at the seat; the Price, the Crossing, the torsor bijection (T16, machine-checked in Appendix A.2), and the 2^m fibre count apply to the wholly odd family $D\tau(U) = \{d : d(\tau x) = \neg d(x) \ \forall x\}$. For the full wholly-odd family on m free orbits, the orientation fibre has exactly 2^m members, hence m independent binary coordinates. For a constrained admissible family A , the exact target-selection price is $H_0(A) = \log_2|A|$ in information, $L(A) = \lceil \log_2|A| \rceil$ in fixed-length bits (§5). A one-bit seat is certified precisely by $|A| = 2$. No counting claim is made for merely non-invariant targets outside this scheme. A register is even when every reading it can

generate is even. This is not a restriction: oracle-invariant proofs are even under oracle attachment; parity-even sieve weights are even under the prime/ P_2 flip; energy-class estimates are even under critical scaling; occurrence arguments are even under multiplicity swap. Every proved method-class barrier on record is the statement that some distinguished register is even under some named flip while the target is odd under it.

One definition earns its place here, because precision requires asking what “outside the register” formally means. Model the register as an observation map $\rho: X \rightarrow Y$ with $\rho(\tau x) = \rho(x)$ — everything the register can read factors through ρ . Then *outside* is no longer a metaphor: a target odd at x cannot factor through ρ at all (T10_factorization, and the existential form T14_wall_factorization: no g with $g \circ \rho = d$ exists, machine-checked). And any added observation a that jointly resolves the target — $d(y) = r(\rho(y), a(y))$ for some readout r — must separate every odd pair: $a(\tau x) \neq a(x)$ (T11_separation, machine-checked). Resolver separation proves only this local, pointwise distinction: a Boolean-valued a may separate every orbit while its alphabet stays $\{0,1\}$. A 2^m alphabet bound concerns a different object — a single global message selecting an arbitrary target from an admissible family A : if $\text{enc}: A \rightarrow S$ and $\text{dec}: S \rightarrow A$ satisfy $\text{dec} \circ \text{enc} = \text{id}_A$, then enc is injective (T12_encoder_injective, machine-checked), so $|S| \geq |A|$. For the full wholly-odd family on m free orbits, $|A| = 2^m$; for a constrained family, the exact global message price is $\log_2 |A|$. The instance count is executed on the exhibited frame in Appendix B (battery K4).

That is the entire setup. The theory follows.

3. The finite frame: the theory in inventory

A frame names data, symmetry, readings, target, and corpus. For involution rows, a reading is even when $r(\tau x) = r(x)$; a Boolean target is odd when $d(\tau x) = 1 - d(x)$, affine in d and linear in $2d - 1$. The coordinate bridge is the wall against inflation: an object domain M reaches the finite theorems only through an equivariant map $B: M \rightarrow X$, a group homomorphism $\varphi: \Gamma_M \rightarrow \Gamma$, and factorization $d_M = d \circ B$. Absent that bridge, a row is a diagnostic map, not an application.

The core stands on three proved legs. Odd-Supply Separation: even generators cannot decide an odd target. Crossing: one even orbit separator beside one odd supply decides the target. Representation Supply: for finite X with group action, invariant observables generate the trivial isotypic component; a target with nontrivial projection is invisible unless supply covers that support. For $\mathbb{Z}/2$, $V = V^+ \oplus V^-$, the invariant side generates V^+ , an odd target lives in V^- , and the minimal missing coordinate is one binary dimension count, not a Shannon claim absent a supplied measure. Proof in five lines. (1) For $\mathbb{Z}/2$ acting on a finite-dimensional space in characteristic zero, the projectors $(1 \pm \tau)/2$ split $V = V^+ \oplus V^-$. (2) Every invariant observable is τ -even, so the algebra the invariant observables generate lies inside V^+ ; equality with all of V^+ is a completeness or orbit-separation assumption, stated where used. (3) An odd target satisfies $d \circ \tau = -d$ in the ± 1 reading, hence lies in V^- . (4) For the permutation representation $V = \mathbb{R}^U$ with $(\tau f)(x) = f(\tau x)$ — the representation this treatise uses, now named; the orbit count is false for an arbitrary representation — V^- is a direct sum of one line per free orbit. A real line carries a continuum of values; the binary reading restricts to sign-valued functions $f: U \rightarrow \{-1, +1\}$, and exactly there the m free orbits give 2^m sign patterns. (5) The missing content is therefore exactly one binary coordinate per free orbit — counted, not measured; the dimension theorem and the Boolean count are related but distinct. $\square$

Table 1 is the development’s theorem ledger: every item counted as new, with its grade, a brief analytic note, and the contribution counted as new object-mathematics. Finite theorems are proved in the development; grammar- and law-level statements are scoped.

Table 1. Master table of new theorem-grade items. Finite theorems are proved here; grammar- and law-level statements are scoped. Corollary 5.2 is a cited calibration and is not counted as new object-mathematics. Receipt scope: the machine-checked development of Appendix A.1 covers the core theorems T1–T15 of this treatise; the companion-numbered items below (4.1–6.4) are proved in the companion development or structural here, as the Grade column states row by row, and Corollary 4.4’s $\mathbb{Z}/2$ content carries the five-line proof of §3.

Item	Grade	Brief analytic note	Contribution counted as new
Theorem 4.1, Odd-Supply Separation	theorem, finite	even generators close under pairing; an odd target flips across τ	named readings cannot decide d

Item	Grade	Brief analytic note	Contribution counted as new
Theorem 4.2, Crossing	theorem, finite	orbit separation leaves only x and τx ; the odd bit kills the τ case	one odd coordinate plus an even separator decides d
Theorem 4.3, Representation Supply ($\mathbb{Z}/2$ isotypic split, Corollary 4.4)	theorem, finite	for the permutation representation, invariant observables land inside V^Γ ; an odd target lives in the per-free-orbit negative lines	sufficiency at fibre refinement needs a completeness hypothesis, stated where used; support coverage is the linear-decoder corollary
Corollary 4.4, One-bit corollary	corollary	for $\mathbb{Z}/2$, $V = V_+ \oplus V_-$ and $\dim V_-$ counts free orbits	one supplied odd coordinate is counted, not a Shannon bit
Theorem 4.5, Provenance Collapse	theorem in G_0 ; structural outside	admit a verified reduction, delete the provenance clause, and test full-signature equivalence	restatement screening is checkable only in a fixed grammar
Theorem 5.1, First-Sufficient-Entry	theorem for named class; structural as law	readings persist; verified constructions enter at dated boundaries	termination changes only when supplied fibers refine target fibers
Theorem 5.3, Seat-Hiding Necessity	theorem	histories equal on visible inputs give the same router output	crossed versus uncrossed requires the seat bit
Proposition 6.1, Unique finest partition	finite theorem	strict positivity makes independence cellwise; common refinement factorizes	unique independent blocks and the count $2^b - 1$ of their nonempty unions
Proposition 6.2, Coalition versus atomic closure	finite theorem	C is independent of A ; $C \oplus A$ is independent of A ; jointly they determine A	coalition closure refuses the spurious atomic domain
Proposition 6.3, Exhibition lemma	analytic on records; structural bridge	a dependent exhibitor cannot be cut from represented blocks	a registration chain follows only under the stated dependence bridge
Proposition 6.4, Determinant readout blindness	finite theorem, scoped	balanced orthogonal triple; Gram $24I_3$; XOR gives one bit	determinant/Gram readouts miss third-order dependence

3.1 The finite-law layer and typed falsification

Four finite instruments guard the reading. Strictly positive finite laws have a unique finest independent partition, with $2^b - 1$ nonempty unions of blocks for b blocks. Coalition closure and atomic closure differ by one bit: pairwise silence can hide a joint bit, exhibited exactly. The exhibition lemma shows when a dependent exhibitor can merge blocks into one component. The balanced orthogonal triple shows determinant/Gram blindness: pairwise factorization with total correlation one bit is invisible to the Gram readout.

Falsification stays typed. An even reading deciding an odd target breaks the theory. A row needing more than one independent orientation coordinate breaks the one-bit claim where proved. A constructor that outputs a restatement breaks the grand form. A crossing entering without provenance breaks the dating law. Each falsifier kills its named face and nothing above it.

4. The Wall

Theorem 1. *If f is even and d is odd at x , then $f \neq d$.*

Proof. $f(\tau x) = f(x)$ by evenness; $d(\tau x) \neq d(x)$ by oddness; so f and d differ at x or at τx . ■

Theorem 2. *No coalition escapes. If f and g are even, every Boolean combination of them is even; hence no Boolean combination of even readings decides an odd target, and by induction no finite repertoire of them does.*

Proof. $\text{op}(f(\tau x), g(\tau x)) = \text{op}(f(x), g(x))$. ■

The register's blindness is not a weakness of current technique. It is a property of the register type. Refining inside the class moves constants and never touches the wall: this is why the prime-gap bound fell from 0.525 toward smaller constants for decades without the parity problem yielding at gap 2, and why three successive complexity barriers each survived every within-class improvement. The wall is a theorem about symmetry, and theorems about symmetry do not negotiate.

Worked exhibit (natural proofs). Take the register to be the natural combinatorial properties: decidable in polynomial time in the truth-table input, true of a large fraction of all functions. The flip is the attachment of a pseudorandom core: under standard cryptographic hardness, every large constructible property reads a function and its pseudorandom twin identically. The target — a circuit lower bound — is odd under that attachment. The Razborov–Rudich theorem is then the Wall typed at this register: no natural proof decides the lower bound while the hardness stands. This exhibit is sketched in the theory's vocabulary; its full formalization in the Lean development is owed, not claimed.

5. The Price

Theorem 3. *A sign vector t that is wholly odd under τ ($t(\tau x) = \neg t(x)$) carries exactly one binary degree of freedom per free orbit: two such vectors agreeing at one point of an orbit agree on the whole orbit. Moreover, wholly odd content has no fixed point: it lives on free orbits only.*

Proof. $t(\tau x) = \neg t(x)$, so the value at x forces the value at τx ; one choice exhausts the orbit. And $\tau x = x$ would force $t(x) = \neg t(x)$, impossible for a bit. ■

The deficit at a blocked decision point is therefore priced exactly: **one bit**. Never two. Never a continuum. Where a problem looks continuous, the one-bit seat is located after the construction, not instead of it: the analyst must still build the modulus, the witness, the operator; but once built, the decision that remains is binary, and what is missing is its orientation, nothing wider.

One premise is named here, at face value, because precision is cheaper than confusion: the one-bit price holds **at a seat**, a decision point where a single free orbit of τ carries the undecided content. The Price's hypothesis is whole anti-invariance on the designated seat-domain U — $\tau^2 = \text{id}$ and τ fixed-point-free on U , $d(\tau x) = \neg d(x)$ for every $x \in U$; at a single seat this reduces to oddness at the seat, and the fibre and torsor statements are theorems about $\text{Dr}(U)$, the wholly anti-invariant family, never about merely non-invariant targets. If a row's obstruction spreads over m unresolved free orbits, the price is m bits, one per orbit, and the theory locates m seats, each crossed by Theorem 4. The price per seat never rises above one, and the seat count is a property of the construction, not a loophole in the pricing. A second precision is owed: the count $|D(U, \tau)| = 2^m$ is the **unconstrained** product case — the full family of anti-invariant targets. If global constraints correlate the orbits (an admissible subfamily $A \subseteq D(U, \tau)$), the total deficit is the logarithmic cardinality price $H_0(A) = \log_2 |A|$, generally smaller and possibly nonintegral. A second price is owed wherever bits are spent rather than counted: the fixed-length code price $L(A) = \min\{\ell \in \mathbb{N} : |A| \leq 2^\ell\} = \lceil \log_2 |A| \rceil$, the length of any binary message selecting a member of A . The two prices agree when $|A|$ is a power of two, both equal one exactly when $|A| = 2$, and an empty A is inconsistent admissibility, not a price. Constraints reduce how many seats remain live; they never raise the price of any seat above one.

6. The Crossing

Theorem 4. *Let the supply s and the target d both be odd at the seat x . Then there is exactly one bit c such that $d = s \oplus c$ on the whole orbit $\{x, \tau x\}$. The reference section is public structure; its unique calibration bit is the sole target-selecting datum — one bit total, never two.*

The hypotheses are the alignment, stated explicitly: the supply must be odd **at the target's seat**, and the calibration must accompany it. No local odd supply decides a target by oddness alone — s odd somewhere else reconstructs nothing here. The corrected crossing sentence, adopted: *relative to a fixed τ -invariant register ρ , no target separating a τ -pair factors through ρ ; on a designated free seat-domain, a wholly anti-invariant target is reconstructed from a wholly anti-invariant reference supply and an orbit-even calibration exactly when $d = s \oplus c$; and for a row whose compatible target family has exactly two members, the calibration has one binary coordinate.*

Proof. On $\{0,1\}$, two bits that both flip under τ either agree or disagree at x ; set $c = d(x) \oplus s(x)$. Then $d = s \oplus c$ at x , and at τx both sides flip, preserving the equality. Uniqueness: $s \oplus c_1 = s \oplus c_2$ forces $c_1 = c_2$. ■

The calibration repays pressure, and the pressure earns a split. There are **two deeds**, and this treatise no longer identifies them:

Deed A — the anchor. The deed fixes the target’s value at one point of the orbit: $b = d(x_0)$. Oddness then fills the orbit for free: $d(\tau x_0) = \neg b$. No supply is required. (**T6a_deed_anchor**, machine-checked.)

Deed B — the supply. An odd carrier s arrives — provenance-clean, denotable in the enlarged register its entry creates, never in the prior one: with ρ_- the τ -invariant observation before arrival, $s(\tau x) = \neg s(x)$ cannot factor through ρ_- ; the dated entry enlarges the register to $\rho_+(x) = (\rho_-(x), s(x))$, through which s factors exactly. The τ -invariance certificate of ρ_- is spent by the enlargement; any later wall identifies a residual symmetry and certifies it anew — together with one naming bit c supplied from outside, and the arrival equation $d = s \oplus c$ lets anything holding (s, c) decide d on the whole orbit. (**T6b_deed_supply**, machine-checked: c is a universally quantified datum in the theorem’s signature, never computed from d .)

The two deeds are different arrivals, and the difference is the answer to the circularity objection. On a single two-point orbit, Deed A alone already decides d — the supply carries no extra information about the target there. What Deed B adds is not information but **addressability**: the register cannot hold b as anything but a fact about one point, while s is an object the register can compute with, transport, and reuse across seats. Where both deeds arrive, consistency forces $c = s(x_0) \oplus b$ — the calibration is the **agreement of the two deeds**, computed from arrived data only (**fT0E_port**, machine-checked). One more theorem completes the picture: the calibration is not merely a bit at the seat but an **even function** — the xor of two wholly odd sections is τ -invariant, so c descends to the orbit space, and the maps $d \mapsto d \oplus s$ and $c \mapsto s \oplus c$ are exact inverses between odd targets and even calibrations (**T8_torsor_forward**, **T8_torsor_backward**, **T9_torsor_inverse_left**, **T9_torsor_inverse_right**, all machine-checked). In the standard vocabulary: $D(U, \tau)$ is a torsor for the even functions, and the reference-section reconstruction is a bijection, not a narrative. For finite frames the fibre count follows: $|D(U, \tau)| = 2^m$, executed on the exhibited frame in Appendix B (battery K4). That one such bit — as anchor or as naming — exists for every blocked row is the One-Cut Hypothesis of Section 8, stated there as conjecture and never as a theorem of the Lean file.

A final precision is owed: **two notions of supply** travel in this theory, and they are priced differently. The *supply-as-section* is a wholly odd function s on the frame — and since $s = d$ is itself such a section, a section can in principle encode the full orientation pattern at once; the per-seat content is that s contributes one odd coordinate per orbit (Theorem 4), no more and no less. The *supply-as-message* is a global code selecting the target from the register-compatible admissible family A : an encoder $\text{enc}: A \rightarrow S$ and decoder $\text{dec}: S \rightarrow A$ with $\text{dec} \circ \text{enc} = \text{id}$. Then enc is injective (**T12_encoder_injective**, machine-checked), so $|S| \geq |A|$ — at least $H_0(A) = \log_2 |A|$ bits of target-selecting information, and when S is a fixed-length binary code of length ℓ , $\ell \geq L(A) = \lceil \log_2 |A| \rceil$, worst case. For the full wholly-odd family on m orbits this is the 2^m bound, executed on the K4 frame. **The one-bit crossing is exactly $|A| = 2$:** not “the final answer is Boolean” — a yes/no output is always one bit by codomain size, which proves nothing — but “the admissible target family relative to the fixed register has exactly two members.” The substantive burden per row is exhibiting that two-member family, and that burden is encoding data, owed per row and exhibited for the parity frame in Section 8.

The crossing is an entry, not a refinement. The supply is odd under the flip (Theorem 1 says nothing even can substitute), it is not measurable with respect to the register per orbit (no map g with $d = g \circ \rho$ exists; the register carries no coordinate of it; at the uniform law, with d wholly odd and ρ invariant, the executed mutual information is zero: every nonempty observation fibre pairs x with τx under opposite labels, so $I(d; \rho) = 0$; for merely seat-odd targets no such assertion is made), and it arrives calibrated by the record of its arrival (Theorem 4’s unique c). Every solved problem on record entered this way: a dated, provenance-clean object the prior register could not denote — the W-functional entering the Ricci-flow corpus in 2002, the Heegner-point height formula entering the BSD corpus in 1986, multidimensional weights entering the sieve corpus in 2013. In each case the routing of the field flipped at the entry and not before. This is a documentary claim about routing, nothing more: the dated object entered the corpus before the resolution, and the field’s proof-paths rerouted at the entry. The treatise claims that shape in the record, not inevitability, and the One-Cut Hypothesis — not this sentence — is what carries the load for rows still open.

Theorems 1, 3 and 4 compose into the theory’s engine: the register cannot see the bit (Wall), the deficit is exactly one bit (Price), and exactly one bit from outside, once it has entered as deed, closes it (Crossing: the calibration is unique given the supply; the arrival itself is no theorem). The full composition is machine-checked (Appendix A, theorem **fT0E_core**).

7. The sign-blind kernel

One might hope a sufficiently clever scalar invariant recovers orientation from magnitude. The verdict kernel of the theory forecloses this. The lock of a three-axis warrant frame is $\det(R) = \lambda^2$, the squared triple product of its rows. Under any reflection D acting by columns, $M \mapsto MD$, the lock satisfies $R(MD) = R(M)$, while $\lambda \mapsto -\lambda$; the displayed determinant $\det(DRD) = \det(R)$ is the same diagonal reflection acting by rows, an identity equally true for the diagonal D used and printed here for cross-check — the two actions are not identified, and the program implements the column action. **The lock is real and the sign is elsewhere**: the determinant is bit-identical under reflection at machine precision, and the orientation it ostensibly encodes is unrecoverable from it. Executed, not asserted (Appendix B, battery K1):

```
|det(DRD) - det(R)| = 0.0000000000000000e+00    (row-action cross-check; the program acts by columns)    λ:
↪ -0.745 ↪ +0.745
```

The lock is invariant under the flip, and invariance is the whole claim. The Gram map $R(M) = MM^T$ is an invariant observation under the reflection, not the even projection: it is nonlinear, it is not idempotent — $R(2I) = 4I$ and $R(R(2I)) = 16I$ — and it forgets more than one reflection, being invariant under every right orthogonal transformation, so its fibres run far larger than $\{M, MD\}$. The valid conclusion is sign-blindness, nothing stronger: no output of this observation separates M from MD . With D the reflection’s diagonal sign matrix, $DD^T = I$, hence $R(MD) = MDD^TM^T = R(M)$ entrywise — and since IEEE negation flips only the sign bit, the two Gram matrices agree not to rounding tolerance but to exactly zero (Appendix B, battery K1b):

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max |R - R(MD)| entrywise = 0.0e+00
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One typing precision is owed: this $\mathbb{Z}/2$ -invariance (sign reversal on a warrant frame) and Section 4’s $\mathbb{Z}/2$ -invariance (the register’s flip on data) are the same anatomy at two different sites, not the same map on the same space. The Gram lock is an invariant observation at the frame level; the register’s blindness is the even projection at the data level — the projector $P_+ = (I + T)/2$ on functions, the only map in this treatise that bears the name. The treatise exhibits the shape twice, typed separately, and identifies neither with the other.

Continuous verdict functionals symmetric under the register’s flip compute magnitudes only: a symmetric functional agrees on M and MD by definition, and M and its reflection sit in opposite orientation components, so no such functional can separate the components; a functional that encoded orientation while staying symmetric would have to smuggle an odd ingredient — the orientation bit itself — and Theorem 1’s anatomy bars odd content from the even register at either site. Truth-signs are a different datatype. This is the exact, executed content of the oldest lesson in the foundations of mathematics, and Section 15 places it.

8. The One-Cut Hypothesis

The theorems price every blocked seat at one bit. Whether the bit always exists is not a theorem, and this treatise does not pretend otherwise.

The One-Cut Hypothesis. *For every blocked row — a Boolean target, a named even register, a dated corpus — there exists a provenance-clean orientation supply whose dated entry decides the target.* The supply is a torsor element, never unique in absolute terms: the Price theorem exhibits 2^m candidates on m free orbits, 64 in the K4 frame of §8.1. **Normalization Postulate.** *Every row of this treatise that asserts a unique orientation supply specifies a target-independent normalization predicate N selecting at most one element of the wholly odd torsor; uniqueness is always relative to N , and Theorem 4’s calibration is computed against whichever element N selects.*

This is the theory’s single conjecture, and it is armed against itself. It dies on named observations, each killing exactly one claim and nothing above it, and the first falsifier is stated operationally, since one must say how an “even reading deciding an odd target” could ever be observed: **typing-firing**. The register of any mapped row carries a published evenness certificate — the paper that proves the method class symmetric under the named flip (relativization for oracle methods, parity-evenness for sieve weights, and so on). A resolution published inside that same certified register, deciding the odd target, is the observation: the certificate and the decision are both in the literature, on the same row, and the conjunction is the kill. No metaphysics is consulted; the firing range is the published record. The six gates stand: a certified even reading deciding an odd target (kills everything); a proved-inert family supplying after all; a blocked row with no pair defeater; a missing supply proved purely continuous; a constructor outputting a restatement; a crossing entering without provenance.

It makes two falsifiable predictions. First: for elliptic curves of analytic order two, a height pairing nondegenerate at order two exists and is constructible under a frozen syntactic contract: the inputs never name the symbols rank or Selmer group — the tabulated Mordell–Weil generators enter as a list of points, their cardinality never named as an input — while the canonical heights, regulator, real period, and analytic order of Sha are admitted in writing as allowed table fields. The exclusion is syntactic and is stated as such: a certified basis carries its rank through its size, and the regulator certifies nondegeneracy of the pairing it comes from, so no independence-from-rank-information claim is made; what is claimed is that the contract’s allowed fields suffice for the computation, and an exhibited obstruction resolves this downward. The first named beachhead is the curve **389.a1** (conductor 389, rank 2, the smallest-conductor rank-two curve in the LMFDB). The contract is executable: every allowed field is tabulated in the same source for the named curve, the predicted order-two nondegenerate height pairing is computable from those tabulated data under the contract, and failure to exhibit it there falsifies the prediction on the spot — one caution carried with it: an unexecuted attempt by an unspecified party does not fire the gate; the contract elects the specified-algorithm claim, not information-sufficiency: a named algorithm, given the frozen data, terminates and returns a pairing matrix H with a checkable certificate $\det(H) > 0$; a certified contradiction of that guarantee refutes the contract, while a failed, imprecise, or unterminated run refutes an implementation and leaves the contract standing: the pairing either exhibits on this curve’s published data or the prediction dies downward. Second: the next crossing among the mapped rows enters through a certificate the prior register had no term to denote — a denotable crossing kills it. A theory that re-describes the landscape cannot be fired by an observation. This one can — at the gates that remain live after their typing: F2, F3, F5, F6, and the two predictions; F1 and F4 are armed against typings and literal readings and cannot, by their stated construction, kill the supply claim.

Under the hypothesis, the barrier literature collapses into one anatomy: seventeen proved barriers across complexity, sieve theory, proof theory, analysis, and topology are one shape — an even register blind to one odd target — and the twenty-three open rows tabulated in the companion volume, including six of the seven Millennium problems, are typed by that anatomy and priced only where (X, τ, ρ, A) is exhibited: **one bit per unresolved orbit at each exhibited row’s decision seat**. The per-row seat counts are encoding data, owed row by row and exhibited below for one row in full; the treatise prices seats, and does not assert $m = 1$ for any row whose seat structure has not been exhibited.

The six enforcement gates of the hypothesis, each naming the exact observation that would break one named claim and nothing above it, are collected in Table 2.

Table 2 | *The six enforcement gates of the One-Cut Hypothesis, each with its named blast radius.* (tier: hypothesis with named falsifier)

Gate	Firing observation	What falls
F1	An even corpus reading decides an odd target	The parity law’s application; everything collapses. Typed as a certificate-consistency check, not an independent test of supply existence: under the printed definitions the observation is impossible, so a reported instance indicts a typing, proof, or implementation — it does not refute the conjecture’s supply claim
F2	A proved-inert named family supplies after all	The restatement-blocked typing of the row
F3	A method-blocked row has no pair defeater	The census leg; the row falls to the diagonal species
F4	A row’s missing supply is provably non-orientation, purely continuous with no Boolean decision seat	The literal one-bit reading of that row only; the hypothesis survives in the after-construction reading, so this gate tests the literal interpretation and cannot kill the surviving form

Gate	Firing observation	What falls
F5	A constructor outputs a restatement	The grand form
F6	A crossing enters without provenance	The dated-entry account of that row

8.1 The worked encoding: the parity row, in full

One encoding is written here, not deferred to a companion table — the row named as sufficient: the prime/ P_2 parity problem of sieve theory.

The frame, exhibited as data. The demand is exact: §2 requires a finite set X and a map $\tau: X \rightarrow X$ with $\tau^2 = \text{id}$ — not two sequences with matching statistics. So the frame is listed, not described. Take $X = \{11, 851, 13, 1273, 17, 437, 19, 2119, 23, 1703, 29, 869\}$: six pairs $\{p, q\}$ with p prime, q semiprime ($\Omega(q) = 2$), and $q \equiv p \pmod{420}$, where $420 = \text{lcm}(1, \dots, 7)$ — hence $q \equiv p \pmod{d}$ for every $d \leq 7$. The flip τ swaps within each pair; $\tau^2 = \text{id}$ and fixed-point-freeness are verified by enumeration, not by prose. The register is every reading factoring through the residue vector ($n \bmod 2, \dots, n \bmod 7$) — the Type I data of the linear sieve at level $D = 7$, where the frame lives. The target is primality. All of it is executed as battery K4 (Appendix B): 27 even residue readings enumerated, every one even on all six pairs, **zero** deciding primality; the fibre of admissible anti-invariant targets counted at $|D(U, \tau)| = 64 = 2^6$, one orientation bit per pair. The row’s priced family is then written, not labeled: $A_{\text{row}} = \{d_0, \neg d_0\}$ with $d_0(n) = [n \text{ is prime}]$ on the frame — two members, both wholly odd, both compatible with the residue record, since τ preserves the residue vector and no reading factoring through it can separate the two worlds. K4 executes the separation proof: of the 27 even residue readings, **zero equal d_0 and zero equal $\neg d_0$** — Theorem 1 applied twice, by enumeration. Hence $|\{d_0, \neg d_0\}| = 2$ by listing — the pair’s non-separation by the 27 residue readings proved by enumeration — and the price of the exhibited family is exactly $\log_2 |A_{\text{row}}| = 1$ **bit**, while the full unconstrained fibre stands at $2^6 = 64$ patterns, **six bits**, the $m = 6$ product count. The two counts price different objects and are never identified: the fibre prices an arbitrary wholly-odd target on the frame; the two-member family prices the row’s exhibited orientation decision. The Wall applies to this frame by citation of Theorem 1, because the frame satisfies Section 2’s hypotheses — verified, not asserted.

The family-level observation, named. Two non-distinguishability statements run through this section on two different domains, and they are separated here, once. The point-level register $\rho: U \rightarrow Y$ observes points of the frame; the wall is the statement that no reading factoring through ρ equals an odd target. The family-level record is a different map on a different domain: $\eta: A \rightarrow Z$ observes candidate targets, and its fibre $A_z = \{d \in A : \eta(d) = z\}$ is the admissible family relative to the record z . For a nonempty fibre the two prices read $H_0(A_z) = \log_2 |A_z|$ in information and $L(A_z) = \lceil \log_2 |A_z| \rceil$ in fixed-length bits, and the one-bit criterion is $|A_z| = 2$, with no global-complement requirement. On this row the record map carries two coordinates, both named inside it: $\eta(d) = (\text{residue-compatibility of } d \text{ with the listed record, } N\text{-class of } d \text{ under the written } N)$. With the observed value z fixed by the residue record and the chosen N -class, the fibre $A_z = \{d \in A : \eta(d) = z\}$ is exactly $\{d_0, \neg d_0\}$ — a fibre of a specified record, not merely a declared subfamily. The coarser reading is stated beside it, at its own price: under residue compatibility alone, every wholly odd candidate on the frame is compatible, because τ preserves the residue vector, so that fibre is the full 64-member wholly-odd family, priced at six bits by the product count. The one-bit count is relative to the specified two-coordinate record; residue data alone do not derive the restriction from the full family, and this treatise does not claim they do. Any row claiming an unresolved target family must likewise say what target-dependent record is available and exhibit its candidates sharing an η -value; declaring a family admissible proves nothing about which record it answers to.

The classical content, typed. The exhibited frame is a finite slice of the parity problem. The classical statement — Type I information alone does not distinguish the parity of $\Omega(n)$ — is precisely the assertion that the sieve register is even under the prime/ P_2 exchange on every such slice; that is the published evenness certificate, and it is why the linear sieve’s optimal bounds land at the parity wall and why no tuning of the weights λ_d inside the class crosses it. The K4 frame is the smallest honest exhibit of the shape; the certificate covers the row.

The target. $d(n) = “n \text{ is prime}”$: odd under τ at every sifted candidate ($\Omega(p) = 1$, $\Omega(p_1 p_2) = 2$ — opposite parities, verified on all six pairs in K4).

The Wall, typed at this row. Every reading in the register is even under τ ; the target is odd; by Theorems 1–2 no sieve weight, and no finite combination of them, decides primality of the sifted elements. On the exhibited frame this is not a re-description but an execution: 27 readings enumerated, none deciding.

The Price, executed by history. The best classical landing is Chen’s theorem (1973): every large even

integer is $p + P_2$. Read against the exhibited shape, the historical record corroborates the anatomy: the sieve register, pushed to its limit, stalls at a binary orientation — prime *or* almost-prime — and fifty years of constant improvements moved θ and never moved that orientation, exactly as the Price predicts for a one-bit seat. The typing is exact: the frame above prices its own exhibited family $A_row = \{d_0, \neg d_0\}$ at one bit — $|\{d_0, \neg d_0\}| = 2$ by listing, the register’s blindness to the pair proved by enumeration; the two-member family is a chosen global flip of the listed primality vector — exhibited, not the fibre of the parity problem, and the residue register does not force the six pair-bits into one locked pattern; on this row the Normalization Postulate’s predicate is written, not assumed — N selects the global flip of the listed primality vector, and uniqueness on this row is relative to that written N ; the Ω parity of Chen’s second summand is **not** a one-bit seat of this frame, and no theorem of this treatise prices the 1+1 row itself, whose seat structure is encoding data owed per row. That the row’s missing content is a provenance-clean orientation supply — one element of the exhibited torsor, unique only relative to a normalization predicate (the Normalization Postulate of §8) — is the One-Cut Hypothesis, conjecture-grade, placed exactly where this treatise has always placed it.

The Crossing, open. The missing supply for 1+1 is one provenance-clean orientation object the Type I register has no term to denote — Type II information beyond the parity-even class, or an object of a kind not yet dated. The exhibited slice stands priced at one bit against its proved two-member family; the row itself stands open, hypothesis-carried, falsifier-armed.

This encoding is instance data: written here by the author, computed by no register, checkable line by line against the published sieve literature. The remaining twenty-two rows carry their certificates in the companion volume under the same gate. The certificates are frozen as of this version’s date and carried under the volume’s integrity hashes; a certificate that changes after the freeze enters as a new dated entry, never as a silent edit. The gate is now stated as a checklist, adopted in force: every application of the finite theorems to a row must exhibit (1) a finite set U ; (2) a free involution τ on it; (3) a fixed invariant register ρ ; (4) an explicitly defined admissible family $A \subseteq \text{Dt}\tau(U)$; (5) a proof that the prior register cannot distinguish members of A ; (6) a count — $|A| = 2$ for a one-bit claim, $|A| = 2^m$ for an m -bit claim — where the one-bit condition is exactly $|A_row| = 2$ — exhibited here as the complement pair $A_row = \{d_0, \neg d_0\}$, the sufficient special case, with membership by listing, not labeling (a row ending in a yes/no question is a codomain fact and proves nothing; the compatible-target family is larger than two until a theorem reduces it); (7) a global encoding model wherever a lower bound on supplied information is asserted. A row missing any item is program, not evidence: the finite theorems reach it only through its bridge, the per-row one-bit claim is valid exactly where $|A_row| = 2$ is exhibited, and the universal-invariant reading rides on the One-Cut Hypothesis alone, exactly where this treatise has always placed it.

9. Provenance, dating, and termination

Terminology, bound once. The register at date t is the §2 observation class extended by every supply that has arrived up to t : readings persist, and each arrived supply appends one independent coordinate to the observation map. Both uses of *register* in this treatise denote this one object at a date — never a bibliography alone, never a physical store.

The bought premise is disciplined by three laws. Provenance Collapse is decidable inside a fixed grammar G_0 : admit a verified conditional reduction, delete its provenance clause, and test equivalence over the shared non-provenance signature; outside G_0 the criterion is structural and never inflated. Dated Inventory separates denotability, proof admission, and verification: readings persist, constructions become readings at entry date, and a token changes only when the dependency hash of new sufficient supply enters the corpus. First-Sufficient-Entry makes termination corpus-relative: the resolution date is the first verified entry whose fibers suffice, not the later retelling.

Seat-Hiding Necessity closes the router escape: without the seat bit, crossed and uncrossed histories are observationally identical. The Perelman control is the crossed calibration, not a premise for open rows: the W -functional and reduced length entered the Ricci-flow corpus as dated constructions, and the corpus rerouted at entry.

9.1 The beachhead and the perimeter

The BSD row is the named beachhead: the corpus already records two dated entries of supply shape, with the order-two exhibition owed. The prediction is constructible and falsifiable: for analytic order two, a height pairing nondegenerate at order two exists under the frozen syntactic contract of Section 8 — inputs never naming the

symbols rank or Selmer group, the generator table entering as a list of points, the regulator and analytic order of Sha admitted in writing as allowed fields; the contract excludes the symbols, not the information a certified basis carries, and claims field-sufficiency, not rank-independence; the smallest named test is curve 389.a1. The perimeter is hard: no object-level theorem is authored; no open row is invoked without bridge and admissible family; the grand form is conditional severance of decision bits, never a bundle of solutions; orientation-after-construction governs continuous-carrier rows.

10. The diagnostic maps

Three maps and one placement table discipline the theory's contact with the open rows. Table 3 fixes six open rows and one crossed control under the coordinate-bridge gate: target bit, object-domain action, halt locus, finite-core status. Table 4 is the SIMS diagnostic at structural grade: for each flow, the settled adjacent fact and the missing monotone supply. Table 5 is the conditional severance map: native flip, even register, missing orientation carrier, what the cut buys, and what it does not prove. No open row invokes the finite theorems until an equivariant bridge is supplied; the maps are diagnostics, not solution claims.

Table 3. Six open rows and one crossed control under Definition 3.2.

Problem	Target bit d	Object-domain action	Halt locus	Finite-core status
P versus NP	equality string	complementation candidate; row locus not supplied	unclassified; no L_H supplied	structural; finite-instance bridges only
Riemann Hypothesis	truth direction	content negation and gauge	reader-barred	structural; no finite X supplied
Yang–Mills	continuum limit exists	strong-coupling alternation	witness-absent	structural; bridge absent
Navier–Stokes	no breakdown	critical scaling $a_* = 3/(2p)$	witness-absent, inert	structural; bridge absent
Hodge	class algebraic	deformation; rigid point	witness-absent	structural; bridge absent
BSD	rank identity	model swap	witness-absent by separation	structural; bridge absent
Poincaré control	noncollapsing	parabolic rescaling	witness-absent, then resolved	crossed calibration; outside the open-row gate

Table 4. SIMS diagnostic, structural grade only.

Row	Flow	Settled adjacent fact / self-similar part	Missing monotone supply
Poincaré control	Ricci flow	κ -solutions classified	W , supplied [61, 48, 65, 64, 63]
Navier–Stokes	equation flow	self-similar blowup excluded [70, 71]	a funnel forcing asymptotic self-similarity
Yang–Mills	RG flow	a -theorem constrains endpoints [68]	force the infrared end to be trivial and gapped
Riemann	de Bruijn–Newman heat flow	adjacent bound $\Lambda \geq 0$ [54]; no self-similar endpoint	force $\Lambda \leq 0$
Hodge; P versus NP	no natural flow	none	no foothold

Table 5. Conditional severance map: six open rows and one crossed control under the Definition 3.2 coordinate-bridge gate (cross-referencing Table 3). No open row invokes the finite theorems until an equivariant bridge is supplied; the table is a conditional severance map, not a solution claim.

Problem	Native flip / defeater	Even register	Missing orientation carrier	What the cut buys	What it does not prove
Riemann Hypothesis	content negation and gauge	functional- equation arguments without Euler product [77]	a self-adjoint operator, constructed from data that do not consult the zeros, whose spectrum is the transformed zeros $t\rho = (\rho - 1/2)/i$ [52]	$t\rho$ real for every ρ forces $\text{Re}(\rho) = 1/2$, in two lines — the imaginary parts alone would not: they are real on or off the line	the operator's existence
P versus NP	complementation as flip (empty fixed set not used as a halt proof)	relativizing, natural, algebrizing argument classes [47, 44, 49]	an obstruction family (multiplicity form; the occurrence form is refuted [53]) separating the two orbit closures at every size	the obstruction witnesses the separation	the obstructions' existence
Navier–Stokes regularity	critical scaling	energy and harmonic- analysis arguments [40]	a data-uniform modulus dominating the alignment defect	the continuation theorem forbids finite-time breakdown	the modulus; the row is orientation- after- construction
Hodge conjecture	deformation; rigid point	semiregular deformation arguments [74, 76, 75]	a semiregular witness for every Hodge class	the deformation argument closes the conditional theorem	the witness; orientation- after- construction
BSD	model swap	rank computations by separation	a rank-identity witness in the Heegner–Euler- system lineage [55, 56]	the analytic-order conditional closes	the witness construction
Yang–Mills existence and mass gap	strong-coupling alternation	continuum-limit arguments	a scale-invariant monotone supply forcing the infrared end trivial and gapped	the flow-row conditional closes	the functional; orientation- after- construction
Poincaré (crossed control)	parabolic rescaling	pre-November- 2002 Ricci-flow reading class [60]	the W-functional and reduced length — supplied 2002 [61, 48]	the crossing is dated and verified	nothing open; this row calibrates the seat

10.1 The depth order

The placement claim closes with the Three-Floor Depth Order, block strength below the formal window. Depth is presupposition-poverty, not power: each deeper floor needs strictly less machinery and therefore holds in strictly more worlds. Receipt rows naming the development's machine-checked files refer to the companion record [1];

Appendices A and B of this paper carry the formal-alone proof and the executed demonstration.

Table 6. *The Three-Floor Depth Order · block strength below the formal window.* The floors are ordered by strictly decreasing presupposition: each deeper floor requires strictly less machinery to state and therefore holds in strictly more worlds. Depth is presupposition-poverty, not power — $\text{req(I)} \supseteq \text{req(II)} \supseteq \text{req(III)}$, three floors total, the Block two presupposition-drops below Gödel’s window.

	Floor I · Gödel Incompleteness (The Downstream Aperture)	Floor II · The Pre-Gödel Bar (Ninth Gate Wall · Unconditional Incompleteness)	Floor III · The Register-Zero Block (Odd-Supply Separation Block Theorem)
Governing result	Gödel 1931 incompleteness	The Ninth Gate wall/self — the typed diagonal, axiom-free: no index carries the diagonal of negation (read); the frame returns to itself, unsettled (compute)	Odd-Supply Separation (Theorems 2.3/4.1), with the Crossing theorem and One-Bit corollary
Minimal machinery	Arithmetic (Robinson Q at minimum), Gödel coding, the diagonal lemma, a provability predicate, consistency hypotheses	Syntax and deeds only: a typed diagonal, actuation, census — no arithmetic, no coding, no consistency hypothesis	Finite data and one involution — no syntax, no arithmetic, no infinite objects
Register location	Inside the ladder: $L2m \subsetneq L1m$, the provability sublayer	Below the ladder, at the Tongue-deed economy — the barzakh of syntax and act	Below the Tongue: pre-syntactic data-and-symmetry, the register every seal presupposes
Engine	The fixed-point-free diagonal — Ground dimension zero	One diagonal, two surfaces: the wall at settled values; the self at the admitted unsettled value	Pure parity: even coalitions under a native involution
Positive content	Truth outruns provability; no consistent recursive system certifies its own consistency	The register cannot settle its own negation diagonal — unconditionally, at every type; the port typed, checkable, content-blind	No even coalition decides an odd target; register-zero is computed and finished; everything above the zero is open by law
Named limit	Bounds the ladder; cannot bound the Ground	Verifies the port, never the crossing; no run confirms a universal	Constrains no open row until an equivariant coordinate bridge is supplied — the Omega Boundary
Standing falsifier	A complete, consistent, recursive extension of Q — one exhibit ends the placement	An index carrying the diagonal of negation — one exhibit breaks the wall	An even readout deciding an odd target, or a provenance-clean entry changing no token of the corpus
Receipt	Metatheoretic proof	GateThesis.lean — fifteen printed theorems, axiom-free, core Lean 4; gate_thesis.f90 kinetic legs	Wallself.lean, RAFGate.lean — block, odd_supply_crosses, seat-hiding; master battery re-runnable

	Floor I · Gödel Incompleteness (The Downstream Aperture)	Floor II · The Pre-Gödel Bar (Ninth Gate Wall · Unconditional Incompleteness)	Floor III · The Register-Zero Block (Odd-Supply Separation Block Theorem)
Grade	[T] theorem	[T] theorem on the wall/self; [Engineering] on the kinetic battery	[T] theorem, finite setting; $\llcorner$ on the computed register-zero
Validity domain	Worlds with arithmetic-rich formal systems	Worlds with syntax and actuation, arithmetic or not — including worlds where Floor I cannot be stated	Worlds with finite sets and a group action — including worlds where Floors I and II cannot even be stated

11. One bit across the barrier landscape

The theory extends across fields without moving its grades. Across complexity, sieve theory, proof theory, analysis, geometry, and topology, proved barriers share one anatomy: the barred register is even under a named flip and the target is odd under it. Relativization bars oracle-invariant arguments from the separation bit. Natural proofs bar large constructible properties from the lower-bound bit while cryptographic hardness stands. Algebrization extends the same evenness to algebraic extension. The sieve parity problem bars parity-even weights from the prime/ P_2 flip. Supercriticality bars energy-class estimates from the sign of the alignment defect. Occurrence refutations bar multiplicity-blind arguments. Independence results sit at the diagonal port, where the system cannot settle its own consistency from within.

This is not a metaphorical unification. It is a typing discipline: every mapped barrier must name its finite set or bridge, its involution or group action, its even register, its odd target, its dated corpus, and its halt locus. A row missing any item is program, not evidence.

Table 7 types the seven Millennium rows under the frame: target bit, native flip or even register, owed supply, gate typing; the Poincaré row is the crossed control, seat-bit certified. Table 8 extends the same typing to sixteen further rows, each with its logical type, native flip or dichotomy, register status, owed supply, and gate typing.

Table 7 | *The seven Millennium rows under the frame.* (tier: structural diagnostic; the Poincaré row is a crossed control)

Row	Target bit	Native flip / even register	Owed supply	Gate typing
P versus NP	separation sign	three barrier classes even under their flips	obstruction family with orientation	exact (structural)
Riemann Hypothesis	the global decision, zeros on the line	Davenport–Heilbronn register (1936)	the operator; orientation-after-construction	exact (structural)
Navier–Stokes	breakdown sign	energy and harmonic-analysis arguments even under critical scaling (Tao 2016)	data-uniform modulus dominating the alignment defect	wall row (F4 gate)
Yang–Mills	mass-gap sign	strong-coupling alternation; continuum-limit arguments	scale-invariant monotone supply, infrared end trivial and gapped	wall row (F4 gate)
Hodge	class sign	integral constructions fail (Atiyah–Hirzebruch 1962; Kollár 1990)	semiregular witness	wall row (F4 gate)

Row	Target bit	Native flip / even register	Owed supply	Gate typing
BSD	rank parity	root number, parity of analytic order	height pairing nondegenerate at order two, built provenance-clean	exact at orders 0, 1 (beachhead); owed at order 2
Poincaré (crossed control)	noncollapsing	parabolic-rescaling reading class	W-functional and reduced length, entered 2002 (Perelman)	exact (crossed; seat-bit certified)

Table 8 | *Sixteen extension rows under the frame, with the logical type of each statement (Π_1 finitely refutable by deed; Π_2 possibly true-but-unprovable; analytic, quantifying over a real-coded continuum). Flag (heuristic) inherits its source’s verification status. (tier: structural diagnostic)*

Row	Logical type	Native flip / dichotomy	Register status	Owed supply	Gate typing
Goldbach	Π_1	prime versus P_2 parity	parity problem named (Selberg)	parity-breaking distribution input	exact (structural)
Twin primes	Π_2	gap-2 parity	Maynard–Tao inertness proved at gap 2	distribution input beyond the proved-inert family	exact (structural)
Legendre	Π_1	prime in $[n^2, (n+1)^2]$ per instance	conditional supplies stop short (Cramér 1936; Baker–Harman–Pintz 2001)	sub-square-root distribution input	exact (structural)
abc	Π_1 per instance	quality $q > 1$ flip	Stewart–Yu plateau named	radical-quality supply above the plateau	exact (structural)
Jacobian	Π_1 per instance	invertibility sign	reduction machinery named	degree-2 certificate or obstruction	exact (structural)
Mersenne primes	Π_2	both polar answers open	none dated on either side	any supply into sparse exponential sequences	pure unbridged target
Odd perfect numbers	Π_1 (nonexistence)	deficient versus abundant sign	abundancy fixed point $I(n) = 2$; floor $N > 10^{1500}$ (Ochem–Rao 2012)	none established	pure unbridged target
Smooth 4D Poincaré	effectively Π_1 -like (heuristic)	exotic S^4 existence versus nonexistence	Whitney trick fails in dimension 4	finite handle-diagram deed or obstruction theory	pure unbridged target
Collatz	Π_2	cycle versus divergence, not an involution	Terras 1976 and Tao 2019 ceilings proved	cycle-exclusion deed or divergence witness	diagonal-port row (F3)

Row	Logical type	Native flip / dichotomy	Register status	Owed supply	Gate typing
Hadwiger	Π_1	minor/complement duality (Steiner 2021)	live register, dated entries 2019 to 2021	linear-bound deed	one bit from closure
Sunflower	Σ_2	C^k versus $(\log k)^k$ bounds	robust-sunflower machinery partial (Alweiss–Lovett–Wu–Zhang 2021)	the constant $C(r)$ itself	one bit from closure
Erdős–Straus	Π_1	exception set empty versus density zero (Vaughan 1970)	modular-identity covering class partial	identity family covering all residue classes	one bit from closure
Beal	Π_1	hyperbolic flip boundary $\sum 1/x < 1$ (Darmon–Granville 1995)	Frey-curve class named	uniform Frey-type parametrization	one bit from closure
Invariant subspace (Hilbert)	analytic	Banach flank negative (Enflo 1976/87; Read 1984), Hilbert flank open	Lomonosov-type register named	cyclic-vector operator or new positivity theorem	wall row: the bridge gate is the bit
Hilbert’s 16th, second part	Σ_2 over reals per n	individual finiteness versus uniform bound	non-accumulation proofs, non-effective	effective cyclicity bound	wall row: the bridge gate is the bit
Lindelöf / Montgomery PCC	analytic	zeros-to-primes duality (Goldston–Montgomery 1987)	moment and decoupling register partial (Bourgain 2017); support $ u < 1$	support-boundary crossing supply	wall row: the bridge gate is the bit

11.1 Halt loci and species

The twenty-three rows are sorted by halt locus, not by field. Boolean-seat rows carry an explicit decision bit and a named even register; there the block shape is exact. Continuous-carrier rows are wall rows under orientation-after-construction: the modulus, witness, operator, or functional must still be built, and the cut orients the built object rather than replacing it. Register-less rows are pure unbridged targets until a coordinate bridge is supplied. Undecidability-flavored rows sit at the diagonal port, where the relevant involution is fixed-point-free and the correct token is halt, not solution.

The species dichotomy is termination typing. Crossed halt: a dated supply entered and the fiber partition reached a deciding refinement. Defeater-absent halt: the row persists because no pair defeater is in the corpus; this is the diagonal species and is not dissolved by rhetoric. The map claims no solution. It prices what each corpus cannot say.

12. Crossed controls, the universal invariant, and the barrier taxonomy

Ten dated crossings are the non-vacuity controls. Heegner points, the W -functional, multidimensional sieve weights, and analogous entries show the same routing: before entry, the register lacks a term; at entry, a provenance-clean construction becomes readable; after entry, proof-paths reroute. The control does not prove open rows will cross. It proves the mechanism has occurred in the record.

The universal one-bit invariant is scoped: where a row exhibits a free involution, a fixed invariant register, an

admissible family $A \subseteq D_\tau(U)$, and a proof that the prior register cannot distinguish members of A , the target-selection price is $\log_2 |A|$ in information, $\lceil \log_2 |A| \rceil$ in fixed-length bits. A one-bit claim is valid exactly at $|A| = 2$: the two members need not be global complements — on two free orbits the family of two sign patterns agreeing on one orbit and differing on the other is wholly odd, two-membered, and complement-free; the complement pair $\{d_0, -d_0\}$ is the sufficient special case this treatise exhibits. A yes/no codomain alone proves nothing. Constraints can reduce the number of live seats; they never raise the price of a seat above one.

The non-vacuity controls are ten dated crossings, each a seat-bit certification: the corpus’s routing flipped only at a dated, provenance-clean entry whose decisive object the dated prior register is not recorded to have denoted. Documentary absence is weaker than a non-expressibility theorem: ‘not recorded to have denoted’ is a historical fact about a corpus, never a finite Wall certificate, and no row below is offered as one. Table 9 certifies them at that documentary grade.

Table 9 | *Ten dated crossings as seat-bit certifications: in each, the corpus’s routing flipped only at a dated, provenance-clean entry whose decisive object the dated prior register is not recorded to have denoted (Thm. 5.3 the conditional instrument).* (tier: documentary certification with the Cor. 5.2 caveat)

Crossing	Blocked corpus	Decisive object	Entry dated	Why the prior register was blind
Perelman, Poincaré and geometrization	Ricci flow could not control collapsing	$\mathcal{W}$ -entropy, reduced length, reduced volume	11 November 2002; March, July 2003	Pre-2002 reading class even under parabolic rescaling; sources absent from the flow literature
Gross–Zagier, BSD rank one	no object encoded L' -data as a point height	Heegner-point height formula	1986	Analytic register carried no height pairing
Kolyvagin, BSD rank identity	Selmer bounds uncontrolled	Euler system of norm-compatible points	1988/1990	No norm-compatible system in corpus
Wiles and Taylor–Wiles, Fermat	no modularity route	deformation rings; the 3-5 switch	1994/1995	Switch not recorded in the prior deformation class; non-denotability conditional (Thm. 5.3)
Green–Tao, primes in progressions	Szemerédi only at positive density	transference principle; GPY pseudorandom majorant	April 2004	No majorant met the pseudorandomness line
Zhang to Maynard, bounded gaps	GPY optimality proved for single-variable weights	multidimensional weights; smooth moduli	2013/2014	The inert family was a theorem, not a practice
Hales–Ferguson, Kepler	score bounds hand-checked	purpose-built score; Flyspeck certificate	1998/2014	Certificate not recorded until formalized; non-denotability conditional (Thm. 5.3)
Appel–Haken, four color	human survey of configurations	machine certificate	1976	Certificate beyond hand-survey readings on the dated record; non-denotability conditional (Thm. 5.3)

Crossing	Blocked corpus	Decisive object	Entry dated	Why the prior register was blind
Tao, discrepancy	positive-density arguments	logarithmic averaging as missing weight	2015	Averaging register not in corpus
Gödel, incompleteness	Hilbert’s program	arithmetized self-referential sentence	1931	Diagonalization required a coding technology invented inside the proof

12.1 The barrier taxonomy mapped

The external barrier literature is mapped into the frame in Table 10: method class ruled out, frame typing, nearest enforcement gate, and the named crossing or escape.

Table 10 | *The barrier literature under the frame: method class ruled out, paper-side typing, nearest enforcement gate, and named crossing or escape.* (tier: structural diagnostic; crossing column documentary)

External barrier	Method class ruled out	Frame typing	Gate	Named crossing or escape
Relativization (Baker–Gill–Solovay 1975)	diagonalization, simulation	reader-barred; common; oracle pair as defeater	F1	IP = PSPACE (Shamir 1990)
Natural proofs (Razborov–Rudich 1997)	constructive large properties	reader-barred; common; deletion-test mirror	F5	Williams ACC bounds (2011)
Algebrization (Aaronson–Wigderson 2008)	all algebrizing arguments	reader-barred; common	F1	Williams code access (2011)
Incompleteness (Gödel 1931)	all proofs within one consistent theory	terrain-empty in-system; diagonal habitat	F3	none internal; new axioms only
Paris–Harrington (1977); Goodstein (Kirby–Paris 1982)	PA-provable Ramsey and termination	theory-side wall; diagonal habitat	F3	provability in stronger systems
Continuum hypothesis (Gödel 1938/40; Cohen 1963)	all of ZFC	terrain-empty for ZFC; diagonal	F3	none
Resolution lower bound (Haken 1985)	polynomial-size Resolution for PHP	reader-barred over Resolution; common	F1	Frege program, open
Bounded-arithmetic unprovability (Ajtai 1994)	constant-depth Frege	reader-barred; theory-side	F3	exponential sharpenings
Naturalized-proofs mirror (Razborov 1995)	bounded-arithmetic lower-bound proofs	deletion-test mirror, theory-side	F5	width machinery; Frege open
Sieve parity problem (Selberg)	classical sieve weights, prime/ P_2 flip	witness-absent; common; parity-even register	F1	none at gap 2; bounded gaps bypass (Maynard 2015)

External barrier	Method class ruled out	Frame typing	Gate	Named crossing or escape
GPY optimality (2005)	single-variable weights below $1/2 + \varepsilon$	restatement-adjacent; proved non-supply of named family	F2	multidimensional weights (Maynard 2015); smooth moduli (Zhang 2014)
Averaged-NS blow-up (Tao 2016)	energy, scaling, symmetry arguments in 3D	witness-absent, inert; energy register even under critical scaling	F4	none; algebraic cancellation required
Occurrence refutation (Bürgisser–Ikenmeyer–Panova 2019)	occurrence arguments	proved non-supply of occurrence family; restatement-adjacent	F2	multiplicity obstructions
Integral Hodge failures (Atiyah–Hirzebruch 1962; Kollár 1990)	integral cycle construction	witness-absent; common; semiregular-deformation register even	F4	rational conjecture survives
Whitney-trick failure, dimension 4	smooth surgery and h-cobordism	reader-barred over smooth-surgery class; witness-absent	F3	topological category only (Freedman 1982)
Conway universality (1972)	uniform decision methods for Collatz-type maps	diagonal habitat, meta-level; theory-side	F3	none known
Montgomery support boundary (1973)	pair correlation beyond tested support	wall row: the bridge gate is the one-bit separation itself	F4	none named

12.2 Standing relations to prior positions

Table 11 closes the engagement ledger: one row per prior position, what it holds, what this paper does with it, the relation word, and the evidence word.

Table 11 | *The standing table: one row per prior position the body engages, relation words as defined above, evidence words from the closed list (documentary, cited, argued); no executed battery is claimed by this paper.* (tier: structural diagnostic)

Prior position	What it holds	What this paper does with it	Relation	Evidence
Relativization (Baker–Gill–Solovay 1975)	Oracle-invariant proofs cannot settle P versus NP	Types it as a reader-barred halt locus with the oracle pair as defeater, gate F1	subsuming	argued
Natural proofs (Razborov–Rudich 1997)	Constructive large properties self-defeat	Types it as the deletion test’s mirror on the theory side, gate F5	subsuming	argued
Algebrization (Aaronson–Wigderson 2008)	Algebrizing arguments cannot separate	Types it reader-barred over the algebrizing class, gate F1	subsuming	argued

Prior position	What it holds	What this paper does with it	Relation	Evidence
Sieve parity problem (Selberg)	Classical weights are parity-even	Reads it as the canonical even register under the prime/ P_2 flip	subsuming	argued
GPY optimality (2005)	Single-variable weights provably stop short	Reads it as a proved non-supply, restatement-adjacent, gate F2	subsuming	argued
Bounded gaps (Zhang 2014; Maynard 2015)	New weight families escaped the wall	Certifies the escape as deed-shaped, outside the inert family	corroborating	documentary
Averaged blow-up (Tao 2016)	Energy-class arguments are exhausted	Types it witness-absent and inert, gate F4, algebraic cancellation owed	subsuming	argued
Occurrence refutation (Bürgisser–Ikenmeyer–Panova 2019)	Occurrence arguments provably fail	Types it as proved non-supply of the occurrence family, gate F2	subsuming	argued
Integral Hodge failures (Atiyah–Hirzebruch 1962; Kollár 1990)	Integral constructions fail in general	Types the row witness-absent with the semiregular-deformation register even, gate F4	subsuming	argued
Independence results (Gödel 1931; Paris–Harrington 1977; Cohen 1963)	No proof inside the system	Places them in the diagonal habitat, theory-side, gate F3, the paper’s one remainder	scoping	argued
Proof complexity bounds (Haken 1985; Ajtai 1994)	Resolution and bounded arithmetic are too weak	Types them reader-barred, theory-side, gates F1 and F3	subsuming	argued
Whitney-trick failure (Casson; Freedman 1982)	Smooth surgery fails in dimension 4	Types SPC4 as a pure unbridged target, reader-barred over the surgery class	scoping	argued
Conway universality (1972)	Collatz-type totality is undecidable in general	Places Collatz at the diagonal port, meta-level, gate F3	scoping	argued
Pair correlation (Montgomery 1973; Goldston–Montgomery 1987)	Support boundary $ u < 1$; zeros-to-primes duality	Names the boundary the field’s own wall; the row is a wall row, the bridge gate the bit	kin	cited

Prior position	What it holds	What this paper does with it	Relation	Evidence
BSD entries (Gross–Zagier 1986; Kolyvagin 1990)	Rank-one identity with Sha finite	Certifies both as dated supply-shaped entries; the order-two pairing is owed	corroborating	documentary
Perelman (2002/03)	Geometrization via the W-functional	Certifies the crossing as the Grand Witness, seat-bit execution	corroborating	documentary
Modularity (Wiles 1995; Taylor–Wiles 1995)	Semistable modularity, hence Fermat	Certifies the entry shape, deformation rings and the 3-5 switch	corroborating	documentary
Primes in progressions (Green–Tao 2008)	Primes contain long progressions	Certifies the transference entry, provenance imported across literatures	corroborating	documentary
Formal proof (Hales–Ferguson 2006; Hales et al. 2017; Appel–Haken 1977)	Machine-checked certificates settle	Certifies non-denotability of the certificate until formalized	corroborating	documentary

13. The kinetic demonstration: existence pays the bit

Why should anything from outside arrive at all? Because outside, things exist, and existence is not free.

The physical root fact: **to exist is to actuate**. Every confined existent carries a strictly positive kinetic floor (the Heisenberg confinement bound, the zero-point energy, the unattainability of absolute zero, each a theorem of physics). Registration is physical: recording one binary digit in any substrate at temperature T dissipates at least $k_B T \ln 2$ of heat. This is Landauer’s principle, experimentally confirmed, and it prices the theory’s interface. Executed (Appendix B, battery K3):

$$Q(\text{one bit registered}) \geq 2.870978885078724\text{e-}21 \text{ J} \quad Q(\text{formal register}) = 0 \text{ J}$$

One precision is owed here, and it strengthens the claim rather than softening it. Landauer’s floor prices **irreversible** registration; reversible computation has no thermodynamic floor. So the interface is stated exactly, as the theory’s **Registration Postulate** — postulate-grade, placed beside the One-Cut Hypothesis, never above it: any computation inside the register may in principle run reversible at zero cost — but a crossing is not a computation inside the register. A crossing is the registration of external fact, an erasure-class event: two live orientations collapse to one in a dated corpus, and that collapse is irreversible. The bit’s price is therefore the erasure floor $k_B T \ln 2$, and no reversible maneuver inside the register avoids it, because the register is precisely what cannot supply the bit. Battery K3 is the invoice arithmetic only: the Landauer number at the conventional bath temperature of 300 K — a chosen reference, not a derived constant; the floor scales linearly in T — and the formal register’s zero, a stipulated model value, not a measurement. The operation the postulate charges is named: the reset $\text{Reset}(0) = \text{Reset}(1) = 0$ of an initially unbiased bit in an isothermal bath at temperature T — two live states collapsed to one, many-to-one and therefore irreversible; for input bias p the ideal charge is $k_B T \cdot h(p)$ with h the binary entropy, and the stipulated $p = 1/2$ is what prices the deed at $\ln 2$, stated here as an assumption of the postulate, not a derived fact. Under these stated erasure assumptions the bath carries the cost: $Q_{\text{bath}} \geq k_B T \ln 2$ and $\Delta S_{\text{bath}} \geq k_B \ln 2$, while the memory’s entropy falls and the total obeys $\Delta S_{\text{total}} \geq 0$ — the inequality belongs to the bath, never to the formal content. The contrast class is acquisition, the copy $(b, 0) \mapsto (b, b)$, which preserves its source and carries no universal floor. A crossing is charged because it is erasure-class — two live orientations collapse to one in a dated corpus — never because information in the

abstract carries temperature. K3 computes the consequent bound given the postulate; it does not verify that the antecedent occurred, and no battery can: whether a physical event is a deed is the postulate's assignment, not a computation's output. Validation stays syntactic and prior: a physically registered false or inconsistent supply pays the same irreversible cost and remains invalid — the price of the deed is no certificate of the deed's content. The postulate is what the kinetic side contributes: the **price** of the deed and its **necessity** — not its arrival per row, which is the One-Cut Hypothesis and remains conjecture. No functor maps the formal into the physical here, and none is needed: the postulate is the interface, declared once, at postulate grade, exactly where the theory places it.

On bridges. Two demands travel under one word, and they are named separately here, once. A **decoder-bridge** — any function built from even readings that recovers the odd orientation — does not exist. Theorem 1 is the proof; battery K2 executes the information arithmetic; the parent corpus's self-reference batteries (no self-settling negation on the two-element algebra; the register that reads itself settles nowhere) close the reflexive escape hatch. Demanding a decoder-bridge is demanding the very thing the wall and the self-battery bar. An **encoding-bridge** — the author-written tuple naming which flip, which register, which seat, which physical two-state device — is data of the instance. It is written by the author, computed by no register, and the wall does not bar naming it. That demand is licit, and Section 8 answers it for the parity row in full. The official gloss, one sentence: *there is no decoder-bridge, ever; encoding-bridges are instance data, and this theory names them row by row.*

The formal register pays exactly zero because it is formal: syntax dissipates nothing. So the one bit the register cannot generate (Theorem 1) is precisely the one bit that only a physical deed can pay for (Landauer). The two constraints meet at a single point and that point is the theory's engine room:

The physical deed is the one-bit conversion interface where the formal shadow becomes actualized reality. The formal side computes the magnitude; existence, from the kinetic side, supplies the sign; and the kinetic side proves the deed's necessity and its price: the only thing the formal side cannot do for itself — pay for one bit — is the one thing existence does by definition.

A proof written down is such a deed. A counterexample exhibited is such a deed. The Heegner point, the W-functional, the machine certificate: each was a physical event in a dated corpus, each paid its bit, and each crossed a wall the register could not. The kinetic demonstration executes this invoice arithmetic and the wall's information arithmetic on the same program: the register carries 0.000000 bits about the odd target, and one supplied bit closes the decision exactly (Appendix B, battery K2).

14. Falsifiable predictions

Two predictions stay exposed. First, the BSD height-pairing prediction remains the named next deed, falsifiable on published curve data. Second, every future crossing is predicted to enter through a certificate the prior even register could not denote; a denotable crossing kills the hypothesis.

15. Placement, and what this theory is

Gödel's incompleteness is not a ceiling over this theory; it is a fixture inside it — as **structural analogy**, stated as analogy. Theorem 1 is a theorem on finite involutive sets; incompleteness needs a provability predicate and a diagonal lemma, and this treatise claims neither. The analogy is exact in shape: the unprovable sentence is a supply the system cannot generate for itself, and Theorem 1 is its finite shadow. Tarski's undefinability reads, in the same analogical sense, as the ladder's confession that truth lives on the far side of the wall: the truth-sign is conserved out of band, exactly as the kernel's blindness (Section 7) executes. Both results stop being limits and become measurements — measured by analogy, labeled by analogy.

What the theory is: a theorem triple with a machine-checked proof, one armed conjecture with six falsifiers and two predictions, one executed kinetic demonstration, and a single sentence of content — an open problem is a blocked register missing one bit, and resolution is the dated arrival of that bit from outside, paid for by the only currency existence recognizes.

What the theory is not: a solution to any named open problem, a claim that the bit must always exist (that is the conjecture, labeled), or a metaphysics smuggled in as mathematics. The wall is proved. The price is proved. The crossing-shape is proved. The rest stands exactly where it stands, at face value, in the open.

The mathematics of the wall is finished. The mathematics of the supply is the work.

The fences stand: no solution claim for any open row; no inflation of hypothesis to theorem; no orientation-instead-of-construction; no strength reading of the depth order Gödel, pre-Gödel bar, Odd-Supply block; no urgency as warrant. The contribution is arrangement, taxonomy, and executed checks.

16. Conclusion

One measurement runs through every section. A blocked row is a register missing one binary coordinate; the coordinate cannot be computed from inside the register; it enters, when it enters, as a dated, provenance-clean deed from outside. The wall, the price, and the crossing shape are theorems, machine-checked in core Lean 4 and executed in Fortran. The One-Cut Hypothesis, that every blocked row admits exactly this entry, is a conjecture, armed with six falsifiers and two predictions, with the Birch and Swinnerton-Dyer order-two exhibition as its named next deed. The maps price twenty-three rows without claiming one. The mathematics of the wall is finished. The mathematics of the supply is the work.

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Appendix A · The formal proof (Lean 4, machine-checked)

The machine-checked theorem content of this treatise, verified by Lean 4.12.0 — core logic only, no libraries, no unproved obligations. At v1.5 the file was re-verified on Lean 4.19.0 and 4.34.0 with the same result. Subsection A.2 below carries its own file and its own receipt. Verification contract: `lean fT0E_Formal_Proof.lean` exits 0 with no output; any weakened hypothesis, deleted conjunct, or inserted `sorry` fails the check. Receipt: **PASS**.

The file proves: `T1_wall` (no even reading decides an odd target), `T2_coalition` (closure under all Boolean joins), `T3_no_fixed_point` (odd content lives on free orbits), `T4_one_bit` (one binary degree of freedom per orbit), `T5_crossing` (exactly one calibration bit decides the target on its orbit: existence and uniqueness), the deed split `T6a_deed_anchor` (oddness propagates one anchored point to the whole orbit; no supply required) and `T6b_deed_supply` (an odd carrier plus one externally supplied naming bit decides the orbit; the naming bit is a universally quantified datum, never computed from the target), their agreement `fT0E_port` (the two deeds force $c = s(x_0) \oplus b$), the full-setup package — `T7_global_wall` (the wall at every point under global oddness), `T8_torsor_forward` and `T8_torsor_backward` (odd sections and even calibrations map into each other), `T9_torsor_inverse_left` and `T9_torsor_inverse_right` (the maps are exact inverses: $D(U, \tau)$ is a torsor for the even functions), `T10_factorization` (an odd target cannot factor through any observation map ρ with $\rho(\tau x) = \rho(x)$: “outside the register” is non-measurability, proved), `T11_separation` (any resolving observation separates every odd pair), `T12_encoder_injective` (an exact global code $\text{dec} \circ \text{enc} = \text{id}$ is injective: the supply alphabet is at least as large as the admissible family; the one-bit crossing is exactly $|A| = 2$), `T13_wholly_odd_is_odd_everywhere` (the two strengths of oddness, typed and never interchanged), `T14_wall_factorization` (no target separating a τ -pair factors through any τ -invariant observation map — the existential form of “outside the register,” proved without finiteness hypotheses), and the Price in bijection form on the canonical frame: `T15_price_bijection_forward` and `T15_price_bijection_backward` (on $\text{Bool} \times Q$ with the flip — whose orbit quotient is Q itself — the wholly anti-invariant targets are in exact bijection with the Boolean functions on the quotient, both round trips proved; the finite cardinal corollary $|\text{Dr}(U)| = 2^{|Q|}$ is executed on the K4 frame and rests, in a mathlib build, on `Fintype.card_congr`, which is outside this core-only file and graded accordingly) — and the engine composition `fT0E_core`. The full source is printed below and verifies with exit 0 and no warnings. This construction continues the machine-checked development of the companion volume (*The Offering Bit*, 2026: `Wallself.lean`, `GateThesis.lean`, `RAFGate.lean`, thirty-three theorems in core Lean 4).

Source: printed in full in Section A.1 of this paper (`fT0E_Formal_Proof.lean`).

A.1 Complete source (Lean 4.12.0, core only)

Verification contract: lean fT0E_Formal_Proof.lean exits 0 with no output and no warnings. Receipt: PASS.
sha256/16: d2258c238be962c8.

```
/-
THE FORMAL THEORY OF EVERYTHING · THE FORMAL PROOF
Lean 4 (core only, no mathlib, zero sorries).

This file proves the complete theorem content of the fT0E:

WALL      (T1) No even reading decides an odd target.
          (T2) No finite coalition of even readings escapes T1.
          (T3) Odd content carries no fixed point.
PRICE     (T4) The odd content of a free orbit is exactly one binary
          degree of freedom: never two, never a continuum.
CROSSING  (T5) One odd supply at the seat, plus exactly one calibration
          bit, decides the target on its orbit; and that calibration
          is unique. One bit missing, one bit supplied, nothing more.
```

What this file does not contain, by its own theorems: the supply itself.
Existence of the supply is the One-Cut Hypothesis, stated as a hypothesis
in the treatise this file accompanies. The wall is proved here. What crosses
it enters by deed, not by derivation.

```
-/

namespace FT0E

variable {α : Type}

/-- A reading f is even under τ when it is constant on τ-orbits. -/
def Even (τ : α → α) (f : α → Bool) : Prop := ∀ x, f (τ x) = f x

/-- A target d is odd under τ when it separates some orbit. -/
def OddAt (τ : α → α) (d : α → Bool) (x : α) : Prop := d (τ x) ≠ d x

/-- A sign vector t is wholly odd under τ when t ∘ τ = ¬t. -/
def WhollyOdd (τ : α → α) (t : α → Bool) : Prop := ∀ x, t (τ x) = !t x

/-- T1 · THE WALL.
An even reading equals no odd target. The register computes every
magnitude and no orientation. -/
theorem T1_wall (τ : α → α) (f d : α → Bool) (x : α)
  (he : Even τ f) (ho : OddAt τ d x) : f ≠ d := by
  intro h
  subst h
  exact ho (he x)

/-- T2 · COALITION CLOSURE.
Evenness is closed under every Boolean join, so no coalition of even
readings escapes the wall. -/
theorem T2_coalition (τ : α → α) (f g d : α → Bool) (x : α)
  (hf : Even τ f) (hg : Even τ g) (op : Bool → Bool → Bool)
  (ho : OddAt τ d x) : (fun y => op (f y) (g y)) ≠ d := by
  apply T1_wall τ _ d x
  · intro y; show op (f (τ y)) (g (τ y)) = op (f y) (g y); rw [hf y, hg y]
  · exact ho

/-- T3 · NO FIXED POINT.
Wholly odd content cannot sit on a fixed point of τ:
t x = ¬t x is a Boolean contradiction. -/
```

```

theorem T3_no_fixed_point ( $\tau : \alpha \rightarrow \alpha$ ) ( $t : \alpha \rightarrow \text{Bool}$ )
  (ho : WhollyOdd  $\tau$  t) ( $x : \alpha$ ) :  $\tau x \neq x$  := by
  intro h
  have h1 :  $t (\tau x) = !t x$  := ho x
  rw [h] at h1
  cases t x <;> simp at h1

/-- T4 · THE PRICE: ONE BIT PER ORBIT.
  Two wholly odd sign vectors agreeing at one point of an orbit agree on
  the whole orbit. The odd content of a free orbit is exhausted by one
  binary choice. -/
theorem T4_one_bit ( $\tau : \alpha \rightarrow \alpha$ ) ( $s t : \alpha \rightarrow \text{Bool}$ )
  (hs : WhollyOdd  $\tau$  s) (ht : WhollyOdd  $\tau$  t)
  ( $x : \alpha$ ) ( $h : s x = t x$ ) :  $s (\tau x) = t (\tau x)$  := by
  rw [hs x, ht x, h]

/-- T5 · THE CROSSING.
  If a supply s and a target d are both odd at the seat x, then there is
  exactly one calibration bit c with  $d = s \text{ XOR } c$  on the whole orbit:
  one supplied orientation bit, one calibration from the dated record,
  and the target is decided. Existence and uniqueness. -/
theorem T5_crossing ( $\tau : \alpha \rightarrow \alpha$ ) ( $s d : \alpha \rightarrow \text{Bool}$ ) ( $x : \alpha$ )
  (hs :  $s (\tau x) = !s x$ ) (hd :  $d (\tau x) = !d x$ ) :
   $\exists c : \text{Bool}, (d x = \text{xor} (s x) c \wedge d (\tau x) = \text{xor} (s (\tau x)) c) \wedge$ 
   $\forall c' : \text{Bool}, (d x = \text{xor} (s x) c' \wedge d (\tau x) = \text{xor} (s (\tau x)) c') \rightarrow c' = c$  := by
  refine (xor (d x) (s x), ?_, ?_)
  · constructor
  · cases s x <;> cases d x <;> rfl
  · rw [hs, hd]
    cases s x <;> cases d x <;> rfl
  · intro c' hc
    obtain (h1, -) := hc
    generalize hsx :  $s x = sv$ 
    generalize hdx :  $d x = dv$ 
    rw [hsx, hdx] at h1
    cases sv <;> cases dv <;> cases c' <;> first | rfl | exact absurd h1 (by decide)

/-- T6a · DEED A – THE ANCHOR (FORGE cycle two: the two deeds, split).
  If d is odd at  $x_0$  and the deed anchors one point ( $d x_0 = b$  – the
  corpus's identification of which sign is which, present in any corpus
  that can pose the decision at all), then d is known on the whole orbit
   $\{x_0, \tau x_0\}$ . NO SUPPLY IS REQUIRED: oddness propagates the anchored bit.
  This is the deed as the arrival of the target's value at one point. -/
theorem T6a_deed_anchor ( $\tau : \alpha \rightarrow \alpha$ ) ( $d : \alpha \rightarrow \text{Bool}$ ) ( $x_0 : \alpha$ ) ( $b : \text{Bool}$ )
  (hd :  $d (\tau x_0) = !d x_0$ ) (anchor :  $d x_0 = b$ ) :
   $d x_0 = b \wedge d (\tau x_0) = !b$  := by
  exact (anchor, by rw [hd, anchor])

/-- T6b · DEED B – THE SUPPLY (FORGE cycle two: the two deeds, split).
  If s is odd at  $x_0$  and c is a naming bit supplied from outside – a
  universally quantified datum, never computed from d – then the arrival
  equation  $d = s \oplus c$ , once it has arrived, lets anything holding the pair
  (s, c) decide d on the whole orbit. This is the deed as the arrival of
  a carrier denotable in the enlarged register  $\rho_+ = (\rho_-, s)$  – never in
  the prior register  $\rho_-$  it was odd against – plus one naming bit. Deeds A and B are
  different deeds; neither is a rewrite of the other inside this file. -/
theorem T6b_deed_supply ( $\tau : \alpha \rightarrow \alpha$ ) ( $s d : \alpha \rightarrow \text{Bool}$ ) ( $x_0 : \alpha$ ) ( $c : \text{Bool}$ )
  (hs :  $s (\tau x_0) = !s x_0$ )
  (arrival :  $d x_0 = \text{xor} (s x_0) c \wedge d (\tau x_0) = \text{xor} (s (\tau x_0)) c$ ) :
   $d x_0 = \text{xor} (s x_0) c \wedge d (\tau x_0) = !\text{xor} (s x_0) c$  := by
  constructor

```

```

· exact arrival.1
· rw [arrival.2, hs]; cases s x0 <;> cases c <;> rfl

/-- THE PORT · the two deeds compose without identifying.
  Where Deed A anchors  $b = d \ x_0$  and Deed B supplies  $s$  odd with naming  $c$ ,
  consistency of the two arrivals forces  $c = s \ x_0 \oplus b$ . The calibration
  is the agreement of the two deeds, computed from arrived data only.
  The missing bit exists as  $b$  (Deed A) or as  $c$  (Deed B); that one such
  bit always exists for every blocked row is the One-Cut Hypothesis,
  stated as conjecture in the treatise – never as a theorem of this file. -/
theorem fTOE_port (_τ : α → α) (s d : α → Bool) (x0 : α) (b c : Bool)
  (anchor : d x0 = b)
  (arrival : ∀ y : α, d y = xor (s y) c) :
  c = xor (s x0) b := by
have h1 := arrival x0
rw [anchor] at h1
generalize hsx : s x0 = sv
rw [hsx] at h1
cases sv <;> cases b <;> cases c <;> first | rfl | exact absurd h1 (by decide)

/-! ## FORGE cycle three · the full-setup package
  The auditors demanded the prose setup be theorem, not preface:
  involution, freeness, global oddness, the observation model, the
  torsor bijection, and resolver separation. Proved here in core Lean. -/

section FullSetup
variable {β γ : Type}

/-- τ is an involution: applying it twice is the identity. -/
def Involution (τ : α → α) : Prop := ∀ x, τ (τ x) = x

/-- τ is fixed-point-free: every orbit is a genuine pair. -/
def FixedPointFree (τ : α → α) : Prop := ∀ x, τ x ≠ x

/-- T7 · THE GLOBAL WALL. Under global oddness, no even reading equals the
  target anywhere it is tested: the wall holds at every point, not one. -/
theorem T7_global_wall (τ : α → α) (f d : α → Bool) (x : α)
  (hf : Even τ f) (hd : WhollyOdd τ d) : f ≠ d := by
intro h
have h1 : f (τ x) = d (τ x) := congrFun h (τ x)
rw [hf x, hd x, congrFun h x] at h1
cases d x <;> simp at h1

/-- T8a · THE TORSOR, FORWARD. The xor of two wholly odd sections is even:
  the calibration against a reference supply descends to the orbit space.
  This is the machine-checked content of "the calibration is a function
  of the orbit, not of the point." -/
theorem T8_torsor_forward (τ : α → α) (s d : α → Bool)
  (hs : WhollyOdd τ s) (hd : WhollyOdd τ d) :
  Even τ (fun x => xor (d x) (s x)) := by
intro x
show xor (d (τ x)) (s (τ x)) = xor (d x) (s x)
rw [hd x, hs x]; cases d x <;> cases s x <;> rfl

/-- T8b · THE TORSOR, BACKWARD. An odd reference xor an even calibration
  is odd: every even calibration pattern lifts to an admissible target. -/
theorem T8_torsor_backward (τ : α → α) (s c : α → Bool)
  (hs : WhollyOdd τ s) (hc : Even τ c) :
  WhollyOdd τ (fun x => xor (s x) (c x)) := by
intro x
show xor (s (τ x)) (c (τ x)) = !xor (s x) (c x)

```

```

rw [hs x, hc x]; cases s x <;> cases c x <;> rfl

/-- T9a · THE TORSOR IS A BIJECTION, FIRST INVERSE. Reference-first:
  s ⊕ (d ⊕ s) = d. Odd sections and even calibrations are in bijection. -/
theorem T9_torsor_inverse_left (s d : α → Bool) :
  (fun x => xor (s x) (xor (d x) (s x))) = d := by
  funext x; cases d x <;> cases s x <;> rfl

/-- T9b · THE TORSOR IS A BIJECTION, SECOND INVERSE. Calibration-first:
  (s ⊕ c) ⊕ s = c. The bijection is exact, not approximate. -/
theorem T9_torsor_inverse_right (s c : α → Bool) :
  (fun x => xor (xor (s x) (c x)) (s x)) = c := by
  funext x; cases s x <;> cases c x <;> rfl

/-- T10 · THE FACTORIZATION OBSTRUCTION. Let the register be an observation
  map ρ with ρ(τx) = ρ(x) – "the register" now means: measurability with
  respect to ρ. A target odd at x cannot factor through ρ. "Outside the
  register" is no longer a metaphor: it is non-measurability, proved. -/
theorem T10_factorization (τ : α → α) (ρ : α → Bool) (g : Bool → Bool)
  (d : α → Bool) (x : α)
  (hp : ρ (τ x) = ρ x) (hd : d (τ x) = !d x) :
  d ≠ fun y => g (ρ y) := by
  intro h
  have h1 : d (τ x) = g (ρ (τ x)) := congrFun h (τ x)
  have h2 : d x = g (ρ x) := congrFun h x
  rw [hp, hd, h2] at h1
  cases hg x : g (ρ x) <;> rw [hg x] at h1 <;> exact absurd h1 (by decide)

/-- T11 · RESOLVER SEPARATION. If an added observation a resolves d jointly
  with ρ – d(y) = r (ρ y) (a y) for some readout r – then a must separate
  every odd pair: a(τx) ≠ a(x). This proves ONLY local, pointwise
  separation: a Bool-valued a may separate every orbit while its alphabet
  stays {0,1}. The 2^m alphabet bound belongs to a different object – a
  single global message selecting one target from an admissible family –
  and is proved by T12's injectivity, never by T11 alone. -/
theorem T11_separation (τ : α → α) (ρ : α → Bool) (a : α → Bool) (r : Bool → Bool)
  (d : α → Bool) (x : α)
  (hp : ρ (τ x) = ρ x) (hd : d (τ x) = !d x)
  (hres : ∀ y, d y = r (ρ y) (a y)) :
  a (τ x) ≠ a x := by
  intro h
  have h1 := hres (τ x)
  have h2 := hres x
  rw [hp, h, hd, h2] at h1
  cases hrv : r (ρ x) (a x) <;> rw [hrv] at h1 <;> exact absurd h1 (by decide)

/-- T12 · THE GLOBAL CODE IS INJECTIVE. If a global message selects the
  target exactly – an encoder Enc and decoder Dec with Dec ∘ Enc = id on
  the admissible family – then Enc is injective, so the supply alphabet
  is at least as large as the family. This is the mechanism behind the
  counting bound: for the full wholly-odd family on m orbits, |S| ≥ 2^m
  (executed on the exhibited frame, battery K4); for an admissible
  subfamily A, |S| ≥ |A|. The one-bit crossing is exactly |A| = 2.
  Grading note: finite-cardinality monotonicity for injections
  (Fintype.card_le_of_injective) is mathlib machinery, outside this
  core-only file; the injectivity is proved here, the instance count is
  executed in K4 (2^6 = 64, pairwise distinct, alphabet ≥ 64). -/
theorem T12_encoder_injective {δ : Type} (Enc : δ → Bool) (Dec : Bool → δ)
  (h : ∀ d, Dec (Enc d) = d) :
  ∀ d1 d2 : δ, Enc d1 = Enc d2 → d1 = d2 := by
  intro d1 d2 hE

```

```

have h1 := h d1
have h2 := h d2
rw [hE] at h1
rw [h2] at h1
exact h1.symm

/-- T14 · THE WALL, FACTORIZATION FORM (the final audit's exact statement).
No target separating a  $\tau$ -pair factors through any  $\tau$ -invariant
observation map: there is no  $g$  with  $g \circ \rho = d$ . Stated without any
finiteness hypothesis – stronger than the requested form, which
assumed [Fintype U]. The proper formulation of "outside the register",
companion to T10 (which quantifies over  $g$  pointwise; T14 closes the
existential). -/
theorem T14_wall_factorization ( $\tau : \alpha \rightarrow \alpha$ ) ( $\rho : \alpha \rightarrow \beta$ ) ( $d : \alpha \rightarrow \text{Bool}$ ) ( $x : \alpha$ )
  (hp :  $\forall z, \rho (\tau z) = \rho z$ ) (hd : OddAt  $\tau d x$ ) :
   $\neg \exists g : \beta \rightarrow \text{Bool}, \forall z, g (\rho z) = d z$  := by
  intro h
  obtain (g, hg) := h
  have h1 :  $d (\tau x) = g (\rho (\tau x))$  := (hg ( $\tau x$ )).symm
  have h2 :  $g (\rho (\tau x)) = g (\rho x)$  := by rw [hp x]
  have h3 :  $g (\rho x) = d x$  := hg x
  exact hd (h1.trans (h2.trans h3))

/-- T13 · THE TWO ODDS, TYPED. Wholly odd implies odd at every point,
checked here. On Bool targets odd-everywhere is equivalent to wholly
odd (two distinct Boolean values are complements), proved as T17 in
fT0E_T16.lean, Appendix A.2. What is false over general targets
is the converse for non-Boolean codomains: a merely non-invariant
target may restrict arbitrarily on other orbits. The torsor (T8, T9)
and the  $2^m$  fibre count apply to the wholly-odd family only; the wall
(T1, T7) needs only oddness at the seat. The two notions are separate
hypotheses in this file and are never interchanged. -/
theorem T13_wholly_odd_is_odd_everywhere ( $\tau : \alpha \rightarrow \alpha$ ) ( $d : \alpha \rightarrow \text{Bool}$ )
  (hd : WhollyOdd  $\tau d$ ) ( $x : \alpha$ ) : OddAt  $\tau d x$  := by
  intro h
  rw [hd x] at h
  cases d x <=> simp at h

/-- The canonical free involution frame:  $\text{Bool} \times Q$  with the flip.
Its orbits are exactly the pairs  $\{(\text{true}, q), (\text{false}, q)\}$ , so its
orbit quotient is  $Q$  itself. -/
def flipF {Q : Type} :  $\text{Bool} \times Q \rightarrow \text{Bool} \times Q$  := fun (b, q) => (!b, q)

/-- T15a · THE PRICE, BIJECTION FORM (canonical frame), FORWARD ROUND TRIP.
On the canonical frame, every wholly anti-invariant target is
reconstructed from its values at the true-section: the map
 $d \mapsto (q \mapsto d (\text{true}, q))$  into  $(Q \rightarrow \text{Bool})$ , followed by
 $c \mapsto ((b, q) \mapsto \text{xor} (c q) (!b))$ , returns  $d$  exactly. Since the orbit
quotient of the canonical frame is  $Q$ , this is one direction of
 $\text{D}\tau(U) \simeq (U/\tau \rightarrow \text{Bool})$ . The exhibited K4 frame is the finite instance
( $Q$  = six seats); the cardinal corollary  $|\text{D}\tau(U)| = 2^{|Q|}$  is executed
there and, in a mathlib build, follows by Fintype.card_congr. -/
theorem T15_price_bijection_forward {Q : Type} (d :  $\text{Bool} \times Q \rightarrow \text{Bool}$ )
  (hd : WhollyOdd flipF d) (b : Bool) (q : Q) :
   $\text{xor} (d (\text{true}, q)) (!b) = d (b, q)$  := by
  have h :  $d (\text{false}, q) = !d (\text{true}, q)$  := hd (true, q)
  cases b
  · generalize ht :  $d (\text{true}, q) = \text{tv}$ 
    generalize hf :  $d (\text{false}, q) = \text{fv}$ 
    rw [ht, hf] at h
    cases tv <=> cases fv <=>

```

```

    first | rfl | exact h | exact h.symm | exact absurd h (by decide)
  · cases d (true, q) <=> rfl

/-- T15b · THE PRICE, BIJECTION FORM (canonical frame), BACKWARD.
  For every  $c : Q \rightarrow \text{Bool}$ , the section  $(b, q) \mapsto \text{xor } (c \ q) \ (!b)$  is wholly
  anti-invariant, and its true-section is  $c$  exactly: the maps are mutual
  inverses between  $D\tau(U)$  and  $(Q \rightarrow \text{Bool})$ . -/
theorem T15_price_bijection_backward {Q : Type} (c : Q → Bool) :
  WhollyOdd flipF (fun (b, q) => xor (c q) (!b)) := by
  intro x
  obtain (b, q) := x
  show xor (c q) (!b) = !xor (c q) (!b)
  cases c q <=> cases b <=> rfl

end FullSetup

/-- The whole theorem content in one line:
  a target odd at its seat is undecidable by the even register (T1, T2),
  missing exactly one bit (T4), and decided by exactly one supplied bit
  plus one calibration (T5). -/
theorem fTOE_core ( $\tau : \alpha \rightarrow \alpha$ ) (s d :  $\alpha \rightarrow \text{Bool}$ ) (x :  $\alpha$ )
  (f g :  $\alpha \rightarrow \text{Bool}$ ) (hf : Even  $\tau$  f) (hg : Even  $\tau$  g) (op :  $\text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}$ )
  (hd : d ( $\tau$  x) = !d x) (hs : s ( $\tau$  x) = !s x) :
  (fun y => op (f y) (g y))  $\neq$  d  $\wedge$ 
   $\exists$  c : Bool, (d x = xor (s x) c  $\wedge$  d ( $\tau$  x) = xor (s ( $\tau$  x)) c)  $\wedge$ 
   $\forall$  c' : Bool, (d x = xor (s x) c'  $\wedge$  d ( $\tau$  x) = xor (s ( $\tau$  x)) c')  $\rightarrow$  c' = c :=
  (T2_coalition  $\tau$  f g d x hf hg op (by intro h; rw [hd] at h; cases d x <=> simp at h),
  T5_crossing  $\tau$  s d x hs hd)

end FT0E

/- Verification contract: this file must pass `lean fTOE_Formal_Proof.lean`
  with no output and exit code 0. Any edit that weakens a hypothesis,
  adds a sorry, or deletes a conjunct must fail that check. -/

```

A.2 The general Price bijection and the Boolean oddness equivalence (T16, T17, machine-checked)

Sections 5 and 8 put the torsor count to load-bearing work: the fibre has 2^m members on m orbits, the supply is a torsor element, and the Normalization Postulate quantifies over the wholly odd torsor. Appendix A.1 machine-checks the one-bit case (T4) and the bijection on the canonical frame (T15). The general bijection and the Boolean oddness equivalence are proved here, in core Lean, in a second file printed whole below. Verification contract: `lean fTOE_T16.lean` exits 0 with no output. The axiom footprint of every theorem in it is asserted inside the file by `#guard_msgs`, which is silent when the kernel's report matches and fails the compile when it does not, so the contract checks the proofs and their dependencies in one command. Receipt: **PASS** on Lean 4.12.0 (commit dc2533473114), Lean 4.19.0 (commit 6caae842e94) and Lean 4.34.0 (commit 293d5d0c0c3f): exit 0, no output, on each. T16 rests on `Quot.sound` alone, T17 on `propext` alone, the reference lemma on no axiom, and nothing on `Classical.choice`. A planted `sorry` in T16 fails the compile at its guard.

Theorem (T16, general Price). *Let U be any type, $\tau : U \rightarrow U$ with $\tau^2 = \text{id}$, and $D\tau(U) = \{d : U \rightarrow \text{Bool} \mid d(\tau x) = \neg d(x)\}$ the wholly anti-invariant family. Fix a reference $d_0 \in D\tau(U)$. No representative selection is needed: $d \oplus d_0$ is even, hence constant on each orbit, so it descends to the quotient. Then the maps*

$$\Phi(d)([x]) = d(x) \oplus d_0(x) : D\tau(U) \rightarrow (U/\tau \rightarrow \mathbb{Z}/2) \quad \Psi(e)(x) = d_0(x) \oplus e([x]) : (U/\tau \rightarrow \mathbb{Z}/2) \rightarrow D\tau(U)$$

are well-defined and mutually inverse, an equivalence $D\tau(U) \simeq (U/\tau \rightarrow \mathbb{Z}/2)$. The XOR action of $(U/\tau \rightarrow \mathbb{Z}/2)$ on $D\tau(U)$ is free and transitive (T8, T9): $D\tau(U)$ is a torsor over the orbit-indexed sign group. The action is canonical and needs no reference; a reference is needed for coordinates, not for the existence of the action.

Proof, four lines. (1) $\Psi(e)$ is anti-invariant: at τx the d_0 summand flips and $e([x])$ does not, so $\Psi(e) \in D\tau(U)$. (2) Φ is well-defined: $d \oplus d_0$ is even, hence constant on each orbit, so its value on $[x]$ needs no chosen representative. (3) $\Psi(\Phi(d))(x) = d_0(x) \oplus d(x) \oplus d_0(x) = d(x)$. (4) $\Phi(\Psi(e))([x]) = d_0(x) \oplus e([x]) \oplus d_0(x) = e([x])$. $\blacksquare$

Scope, exact. The machine-checked statement is wider than the one v1.4 staged, and wider only where the

proof never used the hypothesis. Finiteness is not assumed and decidable equality is not assumed. Fixed-point-freeness is not assumed as a separate hypothesis because the reference carries it: a wholly anti-invariant d_0 cannot exist at a fixed point, since $d_0(x) = \neg d_0(x)$ has no Boolean solution, and `reference_forces_free` proves exactly that. The staged shape, with its fixed-point-free hypothesis present and unconsumed, is `T16_staged_form`. What is not proved here is the finite cardinal corollary: for finite U , $|D\tau(U)| = 2^{|U/\tau|}$ follows from the bijection by a finite cardinality theory that core Lean does not ship, so that step stays at inspection grade, and it is executed at the K4 frame instead.

Theorem (T17, Boolean oddness). *For Bool-valued d , odd at every point is equivalent to wholly odd: $(\forall x, d(\tau x) \neq d(x)) \leftrightarrow (\forall x, d(\tau x) = \neg d(x))$, since two distinct Boolean values are complements.* Machine-checked below for every type U and every map τ , no involution needed. Over non-Boolean codomains the forward direction fails, which is why A.1 keeps the two strengths of oddness as separate hypotheses (T13).

Consistency check, executed: at the K4 frame (12 elements, 6 free orbits) the count is $2^6 = 64$, recomputed in full in Round 1 of the audit; the bijection's cardinality is the count the torsor language has used all along.

Independent witness. The same two theorems, over the same objects, are carried by the master codex of this register, `Lean Codex Seed/TRISDUCTION_Master_Codex_v1.1.4_full.lean` in the public repository `1000sapients/Trisduction` (sha256 `df8d8184f8c15ccfb38056aefc56d727b9796c4f71d32e2f40463027342a2556`, commit `7286ac1`), SECTION 3, as `FTOE.T16_price_bijection` and `FTOE.T17_odd_everywhere_iff_wholly_odd`, each under that file's own axiom guard, with identical dependency lists. The file below needs nothing from that codex.

v1.5 correction, one line: v1.4 staged T16 and T17 with no proof body and a statement that core Lean could not read (`Type*`, `Fintype`); both now carry machine-checked receipts, and the `Fintype` hypothesis is gone because no step of the proof uses it.

Source, `FTOE_T16.lean`, complete:

```

/-
THE FORMAL-ALONE THEORY OF EVERYTHING · APPENDIX A.2 · fTOE_T16.lean
T16, the general Price bijection, and T17, the Boolean oddness equivalence.
Lean 4, core only: no import, no library, no sorry, no axiom declared.

Verification contract: `lean fTOE_T16.lean` exits 0 with no output. The
axiom footprint of every theorem is asserted inside the file by
`#guard_msgs`, which prints nothing when the kernel's report matches and
fails the compile when it does not: T16 rests on Quot.sound alone, T17 on
propext alone, the reference lemma on no axiom, and none on
Classical.choice. The guard strings were generated from the kernel's own
report, never typed.

Scope, exact. T16 is proved for EVERY type  $U$  and EVERY involution  $\tau$  that
admits a wholly anti-invariant reference  $d_0$ . Finiteness is not assumed and
decidable equality is not assumed. Fixed-point-freeness is not assumed
either, because it is not needed as a separate hypothesis: a wholly
anti-invariant  $d_0$  cannot exist at a fixed point, since  $d_0 x = !d_0 x$  has no
Boolean solution, so the reference carries it. The cardinal corollary
 $|D| = 2^{|U/\tau|}$  is NOT proved here; it needs a finite cardinality theory
that core Lean does not ship, and the treatise executes it at the K4 frame
( $2^6 = 64$ ) instead.
-/

namespace FTOE16

/-- The orbit relation of the flip. -/
def OrbitRel {U : Type} (tau : U → U) (x y : U) : Prop := x = y ∨ tau x = y

/-- The wholly anti-invariant targets:  $D(U, \tau)$ . -/
def Dtau {U : Type} (tau : U → U) : Type := { d : U → Bool // ∀ x, d (tau x) = !d x }

/-- A bijection as a four-field record, core only. -/
structure Bij (A B : Type) where
  toFun : A → B

```

```

invFun : B → A
left_inv : ∀ a, invFun (toFun a) = a
right_inv : ∀ b, toFun (invFun b) = b

/-- The calibration of d against the reference d0: d XOR d0. -/
def calib {U : Type} (d0 d : U → Bool) : U → Bool := fun x => xor (d x) (d0 x)

/-- The calibration is even: constant on every orbit. -/
theorem calib_even {U : Type} (tau : U → U) (d0 d : Dtau tau) :
  ∀ x, calib d0.1 d.1 (tau x) = calib d0.1 d.1 x := by
  intro x
  show xor (d.1 (tau x)) (d0.1 (tau x)) = xor (d.1 x) (d0.1 x)
  rw [d.2 x, d0.2 x]
  cases d.1 x <=> cases d0.1 x <=> rfl

/-- Phi: descend the even calibration to the orbit quotient. No representative is chosen. -/
def priceForward {U : Type} (tau : U → U) (d0 : Dtau tau) (d : Dtau tau) :
  Quot (OrbitRel tau) → Bool :=
  Quot.lift (calib d0.1 d.1) (by
    intro a b hab
    cases hab with
    | inl h => rw [h]
    | inr h => rw [← h]; exact (calib_even tau d0 d a).symm)

/-- Psi: lift a function on the quotient back to a wholly anti-invariant target. -/
def priceBackward {U : Type} (tau : U → U) (hinv : ∀ x, tau (tau x) = x) (d0 : Dtau tau)
  (e : Quot (OrbitRel tau) → Bool) : Dtau tau :=
  (fun x => xor (d0.1 x) (e (Quot.mk (OrbitRel tau) x))), by
  intro x
  show xor (d0.1 (tau x)) (e (Quot.mk (OrbitRel tau) (tau x)))
    = !xor (d0.1 x) (e (Quot.mk (OrbitRel tau) x))
  have hq : Quot.mk (OrbitRel tau) (tau x) = Quot.mk (OrbitRel tau) x :=
    Quot.sound (Or.inr (hinv x))
  rw [hq, d0.2 x]
  cases d0.1 x <=> cases e (Quot.mk (OrbitRel tau) x) <=> rfl

/-- T16 · THE GENERAL PRICE BIJECTION. For every type U, every involution tau and
  every wholly anti-invariant reference d0: D(U, tau) is in bijection with the
  Boolean functions on the orbit quotient U/tau, both round trips proved. -/
theorem T16_price_bijection {U : Type} (tau : U → U) (hinv : ∀ x, tau (tau x) = x)
  (d0 : Dtau tau) : Nonempty (Bij (Dtau tau) (Quot (OrbitRel tau) → Bool)) :=
  ({ toFun := priceForward tau d0
    invFun := priceBackward tau hinv d0
    left_inv := by
      intro d
      apply Subtype.ext
      funext x
      show xor (d0.1 x) (xor (d.1 x) (d0.1 x)) = d.1 x
      cases d.1 x <=> cases d0.1 x <=> rfl
    right_inv := by
      intro e
      funext q
      induction q using Quot.ind with
      | _ x =>
        show xor (xor (d0.1 x) (e (Quot.mk (OrbitRel tau) x))) (d0.1 x)
          = e (Quot.mk (OrbitRel tau) x)
        generalize e (Quot.mk (OrbitRel tau) x) = b
        generalize d0.1 x = a
        cases a <=> cases b <=> rfl })

/-- The reference forces fixed-point-freeness: no wholly anti-invariant target

```

```

    exists at a fixed point. This is why T16 needs no separate hypothesis. -/
theorem reference_forces_free {U : Type} (tau : U → U) (d0 : Dtau tau) :
  ∀ x, tau x ≠ x := by
  intro x h
  have h1 := d0.2 x
  rw [h] at h1
  cases hx : d0.1 x <=> rw [hx] at h1 <=> exact absurd h1 (by decide)

/-- T16 in the shape Appendix A.2 of v1.4 staged, with its fixed-point-free
    hypothesis, which the proof does not consume. -/
theorem T16_staged_form {U : Type} (tau : U → U) (h2 : ∀ x, tau (tau x) = x)
  (_hf : ∀ x, tau x ≠ x) (d0 : Dtau tau) :
  Nonempty (Bij (Dtau tau) (Quot (fun x y => x = y ∨ tau x = y) → Bool)) :=
  T16_price_bijection tau h2 d0

/-- T17 · THE BOOLEAN ODDNESS EQUIVALENCE. On Bool-valued targets, odd at every
    point is equivalent to wholly odd: two distinct Boolean values are complements. -/
theorem T17_odd_everywhere_iff_wholly_odd {U : Type} (tau : U → U) (d : U → Bool) :
  (∀ x, d (tau x) ≠ d x) ↔ (∀ x, d (tau x) = !d x) := by
  constructor
  · intro h x
    have hx := h x
    cases hd : d x <=> cases ht : d (tau x) <=> simp_all
  · intro h x hc
    have hx := h x
    rw [hc] at hx
    cases hd : d x <=> rw [hd] at hx <=> exact absurd hx (by decide)

end FT0E16

/-- info: 'FT0E16.T16_price_bijection' depends on axioms: [Quot.sound] -/
#guard_msgs in #print axioms FT0E16.T16_price_bijection
/-- info: 'FT0E16.T16_staged_form' depends on axioms: [Quot.sound] -/
#guard_msgs in #print axioms FT0E16.T16_staged_form
/-- info: 'FT0E16.reference_forces_free' does not depend on any axioms -/
#guard_msgs in #print axioms FT0E16.reference_forces_free
/-- info: 'FT0E16.T17_odd_everywhere_iff_wholly_odd' depends on axioms: [propext] -/
#guard_msgs in #print axioms FT0E16.T17_odd_everywhere_iff_wholly_odd

```

Appendix B · The kinetic demonstration (Fortran, executed)

ft0E_Kinetic_Demonstration.f90 — five batteries, fixed build (gfortran -std=f2018 -O2 -fno-fast-math -ffp-contract=off), oracle halt-on-fail:

- **K1 · The sign-blind kernel.** $\det(R) = \lambda^2$ to $3.3\text{e-}16$ — the printed residual $|\det(R) - \lambda^2|$ is diagnostic display; the battery's pass condition tests exactly two things: reflection invariance of R and sign reversal of $\det(M)$, and it passes or fails on those, not on the residual. Under reflection the lock is bit-identical while λ flips sign. Magnitude in the register; direction not in the register.
- **K1b · Kernel invariance, numerical.** Since $DD^T = I$, $R(MD) = R(M)$ entrywise; IEEE negation flips only the sign bit, so the two Gram matrices agree to $0.0\text{e}+00$ difference — a numerical equality test, as floating-point comparison is (signed zero compares equal to zero); no raw bit-pattern comparison is claimed. The kernel is invariant under the frame's flip, executed to zero difference.
- **K2 · The one-bit wall.** Exact enumeration on the XOR law: the even register carries 0.000000 bits about the odd target; the coalition of register plus one supplied bit decides the target with zero residual entropy. The deficit is one bit; the crossing is one bit. Precision note: K2 is the minimal exhibit at the seat's symmetric prior — mutual information measured at the uniform distribution the flip itself induces. Theorem 1 is stronger than any such measurement: it is about decision (pointwise equality of functions), not correlation, and no reweighting of priors moves a theorem.

- **K3 · The deed's price.** Landauer floor for registering one bit at the conventional 300 K reference temperature: $Q \geq 2.87\text{e-}21$ J; the formal register's payment is the stipulated model value $Q_{\text{formal}} = 0$ — assigned by definition of the formal layer, and the battery's zero test verifies the assignment, not a physical measurement. The deed pays what the register cannot.
- **K4 · The parity frame, exhibited.** Six listed prime/semiprime pairs with matching residues mod 420 = lcm(1..7): (11, 851 = 23 · 37), (13, 1273 = 19 · 67), (17, 437 = 19 · 23), (19, 2119 = 13 · 163), (23, 1703 = 13 · 131), (29, 869 = 11 · 79). Verified by enumeration in the run: $\tau^2 = \text{id}$, fixed-point-free, primality odd on all six pairs, residue match mod d for every $d \leq 7$; 27 even residue readings enumerated, every one even on all six pairs, zero deciding primality; fibre count $|D(U, \tau)| = 64 = 2^6$, one orientation bit per pair; the 64 targets verified pairwise distinct, so any exact global selector needs an alphabet of at least 64 values — at least 6 bits (T12's counting face). The two-member admissible family $A_{\text{row}} = \{d_0, \neg d_0\}$, $d_0 = \text{primality}$, is exhibited as data: both members wholly odd, and of the 27 even residue readings zero equal d_0 and zero equal $\neg d_0$ — the register cannot separate the family, $|A_{\text{row}}| = 2$ by listing, the non-separation of its two members proved by enumeration, the exhibited family's price $\log_2 |A_{\text{row}}| = 1$ bit, never identified with the unconstrained six-bit fibre count. Theorem 1 applies to the frame by citation.

Receipt, three receipts kept separate and never merged: (i) hand verification of the printed prose and derivations; (ii) execution of an arithmetic-identical mirror, gfortran unavailable in the checking environment — **all batteries pass, exit 0**; (iii) compilation and execution of the printed Fortran source under the named flags — not performed in the checking environment, owed as an external receipt, and not claimed here.

External corroboration, on record and weighted as such: an independent replication on a separate substrate recomputed the K3 arithmetic from the sealed description to all sixteen printed digits ($Q = 2.870978885078724\text{e-}21$ J at 300 K). This is the first external replication of the demonstration's arithmetic. It is corroboration-grade only — replication of a computation, not endorsement of the conjecture — and it carries no social weight in the theory's warrant, which rests on the theorems and the executed batteries alone.

Parent batteries, on record in the same repository: `TRISDUCTION_Core_Thesis_Fortran_v2_4_0.f90` (the core thesis executed) and `RA_TOE_Thesis_Fortran_v2_0_0.f90` (the existence root executed), at `protocols/Executable Thesis/`.

The wall is a theorem. The price is a theorem. The crossing-shape is a theorem. The supply is a deed. That is the whole theory.

B.1 Complete source (Fortran 2018)

Build: `gfortran -std=f2018 -O2 -fno-fast-math -ffp-contract=off`. Oracle halt-on-fail: a binary that reaches its final line has passed. Receipt (arithmetic-identical mirror, gcc): all five batteries pass, exit 0. sha256/16: 05eea8b626a9e63c.

```
!=====
!  fTOE . THE KINETIC DEMONSTRATION
!  THE PHYSICAL DEED IS THE ONE-BIT CONVERSION INTERFACE WHERE THE FORMAL
!  SHADOW BECOMES ACTUALIZED REALITY
!
!  The formal side is proved elsewhere (fTOE_Formal_Proof.lean: the wall,
!  the price, the crossing, machine-checked). This file executes the other
!  half of the theory: the kinetic face. Existence is actuation, and every
!  actuation of one bit carries a non-zero thermodynamic price the formal
!  register cannot pay. Three batteries, computed in this run, oracle
!  halt-on-fail:
!
!  K1 THE SIGN-BLIND KERNEL. The verdict lock is  $\det(R) = \lambda^2$ . Under
!  any reflection the lock is bit-identical and the sign of  $\lambda$ 
!  flips. The formal register carries magnitude; direction is not in it.
!
!  K1b KERNEL = EVEN PROJECTION.  $D D^T = I$ , so  $R(MD) = R(M)$  entrywise.
```

```

! IEEE negation is a sign-bit flip, hence the two Gram matrices agree
! to exactly 0.0. The kernel is flip-invariant, zero-difference numerical.
!
! K2 THE ONE-BIT WALL. On the XOR law, exact enumeration: the even
! register carries zero bits about the odd target; one supplied odd
! bit at the seat, with one calibration, decides the target exactly.
! The deficit is one bit wide. The crossing is one bit deep.
!
! K3 THE DEED'S PRICE. Registering one binary orientation in any physical
! substrate at temperature T dissipates  $Q \geq k_B T \ln 2$  (Landauer).
! The formal register pays exactly 0. The deed pays the floor. The
! conversion of formal possibility into actualized fact is a
! thermodynamic transaction, and this battery prints its invoice.
!
! K4 THE PARITY FRAME, EXHIBITED AS DATA. Six listed prime/semiprime
! pairs with matching residues mod  $\text{lcm}(1..7)=420$ ; tau swaps within
! pairs. Involution, freeness, oddness, register evenness, the wall
! on every pair, the fibre count  $2^6 = 64$ , and the encoder bound
! (targets pairwise distinct; exact-selector alphabet  $\geq 64$ ) – all
! verified by enumeration in the run. T1 applies by citation.
!
! Parent batteries, audited and on record in the same repository:
! protocols/Executable Thesis/TRISDUCTION_Core_Thesis_Fortran_v2_4_0.f90
! protocols/Executable Thesis/RA_T0E_Thesis_Fortran_v2_0_0.f90
!
! BUILD (sealed): gfortran -std=f2018 -O2 -fno-fast-math -ffp-contract=off
! A binary that reaches its final line has passed. Nothing is narrated;
! every figure is computed in the run. Receipt, not essence.
! Delta-M = 0. No new mathematics is authored by this file.
!=====

```

```

program ftoc_kinetic_demonstration
  use iso_fortran_env, only: int64
  implicit none
  integer, parameter :: dp = selected_real_kind(15, 307)
  real(dp), parameter :: KB = 1.380649e-23_dp      ! J/K, exact, SI 2019
  real(dp), parameter :: T_ROOM = 300.0_dp         ! K, demonstration bath
  integer :: failures
  real(dp) :: M(3,3), R(3,3), Mr(3,3), Rr(3,3)
  real(dp) :: lam, lamr, detR, detRr, maxdiff
  real(dp) :: I_even, H_joint, Q_bit, Q_formal

  failures = 0

  ! ----- K1 · THE SIGN-BLIND KERNEL -----
  M(1,:) = [ 0.3_dp, 0.7_dp, -0.2_dp]
  M(2,:) = [ 0.5_dp, -0.4_dp, 0.8_dp]
  M(3,:) = [-0.6_dp, 0.1_dp, 0.9_dp]
  R = matmul(M, transpose(M))
  lam = det3(M)
  detR = det3(R)
  Mr = M; Mr(:,3) = -Mr(:,3)      ! reflection D
  Rr = matmul(Mr, transpose(Mr))
  lamr = det3(Mr)
  detRr = det3(Rr)

  write(*, '(A)') 'K1 THE SIGN-BLIND KERNEL'
  write(*, '(A,ES22.15)') ' lambda          = ', lam
  write(*, '(A,ES22.15)') ' det(R)          = ', detR
  write(*, '(A,ES22.15)') ' |det(R) - lambda^2| = ', abs(detR - lam*lam)
  write(*, '(A,ES22.15)') ' reflected lambda = ', lamr

```

```

write(*,'(A,ES22.15)') ' |det(DRD) - det(R)| (row-action cross-check) = ', abs(detRr - detR)
if (.not. (abs(detRr - detR) < 1.0d-12 .and. abs(lamr + lam) < 1.0d-12)) then
    failures = failures + 1
    write(*,'(A)') ' K1: FAIL'
else
    write(*,'(A)') ' K1: lock invariant under reflection; sign unrecoverable. PASS'
end if

! ----- K1b · KERNEL = EVEN PROJECTION -----
! D is a diagonal sign involution,  $D D^T = I$ , hence
!  $R(MD) = M D D^T M^T = R(M)$  entrywise. IEEE negation flips only the
! sign bit, so  $(-a)(-b) = a*b$  bitwise: the two Gram matrices agree to
! exactly 0.0. The kernel is not merely invariant; it IS the tau-even
! projection of the frame, computed bit-exact.
maxdiff = maxval(abs(R - Rr))
write(*,'(A)') 'K1b KERNEL = EVEN PROJECTION (entrywise, bit-exact)'
write(*,'(A,ES9.1)') ' max |R - R(MD)| entrywise = ', maxdiff
if (.not. (maxdiff == 0.0_dp)) then
    failures = failures + 1
    write(*,'(A)') ' K1b: FAIL'
else
    write(*,'(A)') ' K1b: kernel flip-invariant, zero-difference numerical. PASS'
end if

! ----- K2 · THE ONE-BIT WALL -----
call xor_wall(I_even, H_joint)
write(*,'(A)') 'K2 THE ONE-BIT WALL (XOR law, exact enumeration, uniform)'
write(*,'(A,F10.6,A)') ' I(even register ; odd target) = ', I_even, ' bits'
write(*,'(A,F10.6,A)') ' H(target | register + one supply) = ', H_joint, ' bits'
if (.not. (abs(I_even) < 1.0d-12 .and. abs(H_joint) < 1.0d-12)) then
    failures = failures + 1
    write(*,'(A)') ' K2: FAIL'
else
    write(*,'(A)') ' K2: deficit exactly one bit; one supplied bit closes it. PASS'
end if

! ----- K3 · THE DEED'S PRICE -----
Q_bit = KB * T_ROOM * log(2.0_dp) ! Landauer floor for one bit at 300 K
Q_formal = 0.0_dp ! stipulated model value: the formal register pays by definition
write(*,'(A)') 'K3 THE DEED'S PRICE (Landauer floor, T = 300 K)'
write(*,'(A,ES22.15,A)') ' Q(one bit registered) >= ', Q_bit, ' J'
write(*,'(A,ES22.15,A)') ' Q(formal register) = ', Q_formal, ' J'
if (.not. (Q_bit > 0.0_dp .and. Q_formal == 0.0_dp)) then
    failures = failures + 1
    write(*,'(A)') ' K3: FAIL'
else
    write(*,'(A)') ' K3: the deed pays what the register cannot. PASS'
end if

! ----- K4 · THE PARITY FRAME, EXHIBITED -----
! X = six listed pairs {p, q}: p prime, q semiprime ( $\Omega(q) = 2$ ),
!  $q \equiv p \pmod{\text{lcm}(1..7) = 420}$ , hence  $q \equiv p \pmod{d}$  for every  $d \leq 7$ .
! tau swaps within each pair; the target is primality. The register is
! every reading factoring through the residue vector ( $n \bmod 2..7$ ): the
! Type I data of the linear sieve at level  $D = 7$ . Everything is verified
! by enumeration in this run: involution, freeness, oddness, register
! evenness, the wall on every pair, and the fibre count  $2^6$ .
block
    integer(int64) :: P(6), Q(6), X(12), tx, ttx, fibre
    integer :: i, mm, aa, n_even, n_deciding, fp, fq
    integer :: t1, t2, pt, pair, flip, v1, v2

```

```

integer :: n_eq0, n_eq1, dp, dq
logical :: ok4, even_r, decides_r, distinct_ok, same_t, eq0, eq1
P = [11_int64, 13_int64, 17_int64, 19_int64, 23_int64, 29_int64]
Q = [851_int64, 1273_int64, 437_int64, 2119_int64, 1703_int64, 869_int64]
do i = 1, 6
    X(2*i-1) = P(i); X(2*i) = Q(i)
end do
ok4 = .true.
do i = 1, 12
    ! involution + freeness
    tx = X(ieor(i-1,1)+1); ttx = X(ieor(ieor(i-1,1),1)+1)
    if (ttx /= X(i) .or. tx == X(i)) ok4 = .false.
end do
do i = 1, 6
    ! oddness + residue match
    if (.not. isprime_l(P(i)) .or. isprime_l(Q(i)) &
        .or. omega_l(Q(i)) /= 2) ok4 = .false.
    do mm = 2, 7
        if (mod(P(i),int(mm,int64)) /= mod(Q(i),int(mm,int64))) ok4 = .false.
    end do
    if (mod(P(i),420_int64) /= mod(Q(i),420_int64)) ok4 = .false.
end do
n_even = 0; n_deciding = 0
do mm = 2, 7
    ! register evenness + wall
    do aa = 0, mm-1
        even_r = .true.; decides_r = .true.
        do i = 1, 6
            fp = 0; if (mod(P(i),int(mm,int64)) == int(aa,int64)) fp = 1
            fq = 0; if (mod(Q(i),int(mm,int64)) == int(aa,int64)) fq = 1
            if (fp /= fq) even_r = .false.
            if ((fp == 1) .neqv. isprime_l(P(i))) decides_r = .false.
            if ((fq == 1) .neqv. isprime_l(Q(i))) decides_r = .false.
        end do
        if (even_r) then
            n_even = n_even + 1
            if (decides_r) n_deciding = n_deciding + 1
        end if
    end do
end do
if (n_even == 0 .or. n_deciding /= 0) ok4 = .false.
fibre = 1
do i = 1, 6
    fibre = fibre*2
end do
! encoder bound (T12's counting face): the 2^6 anti-invariant targets
! are pairwise distinct as Boolean vectors on the frame, so any exact
! global selector needs an alphabet of at least 64 values.
distinct_ok = .true.
do t1 = 0, 63
    do t2 = t1+1, 63
        same_t = .true.
        do pt = 0, 11
            pair = pt/2; flip = mod(pt,2)
            v1 = ieor(int(ibits(t1,pair,1)), flip)
            v2 = ieor(int(ibits(t2,pair,1)), flip)
            if (v1 /= v2) same_t = .false.
        end do
        if (same_t) distinct_ok = .false.
    end do
end do
if (.not. distinct_ok) ok4 = .false.
! K4 continuation (FORGE v7.1): the two-member admissible family
! A_row = {d0, 1-d0}, d0 = primality on the frame, exhibited as data.

```

```

! Both members are wholly odd (d0 odd on every pair, checked above;
! the complement inherits oddness pointwise). The register cannot
! separate the family: no even residue reading equals either member
! (Theorem 1 applied twice, executed here by enumeration). Hence
! |A_row| = 2 is by listing, not labeling, and the frame's priced row
! claim is exactly one bit (log2|A_row|) -- against the full
! unconstrained fibre 64 = 2^6, the m = 6 product count (six bits).
n_eq0 = 0; n_eq1 = 0
do mm = 2, 7
  do aa = 0, mm-1
    even_r = .true.; eq0 = .true.; eq1 = .true.
    do i = 1, 6
      fp = 0; if (mod(P(i),int(mm,int64)) == int(aa,int64)) fp = 1
      fq = 0; if (mod(Q(i),int(mm,int64)) == int(aa,int64)) fq = 1
      if (fp /= fq) even_r = .false.
      dp = 0; if (isprime_l(P(i))) dp = 1
      dq = 0; if (isprime_l(Q(i))) dq = 1
      if (fp /= dp .or. fq /= dq) eq0 = .false.
      if (fp /= 1-dp .or. fq /= 1-dq) eq1 = .false.
    end do
    if (even_r) then
      if (eq0) n_eq0 = n_eq0 + 1
      if (eq1) n_eq1 = n_eq1 + 1
    end if
  end do
end do
if (n_eq0 /= 0 .or. n_eq1 /= 0) ok4 = .false.
write(*,'(A)') 'K4 THE PARITY FRAME, EXHIBITED (six listed pairs, mod 420)'
write(*,'(A)') ' pairs: (11,851=23x37) (13,1273=19x67) (17,437=19x23)'
write(*,'(A)') '      (19,2119=13x163) (23,1703=13x131) (29,869=11x79)'
write(*,'(A,I0,A,I0)') ' even residue readings enumerated = ', n_even, &
' ; deciding readings = ', n_deciding
write(*,'(A,I0,A)') ' |D(U,tau)| = ', fibre, &
' = 2^6 orientation patterns on the frame (six bits, unconstrained)'
if (distinct_ok) then
  write(*,'(A,I0,A)') ' targets pairwise distinct: verified by &
&enumeration ; exact-selector alphabet >= ', fibre, ' (>= 6 bits)'
else
  write(*,'(A)') ' targets pairwise distinct: VIOLATION'
end if
write(*,'(A)') ' A_row = {d0, 1-d0} exhibited as data: d0 = &
&primality, both members wholly odd'
write(*,'(A,I0,A,I0)') ' even readings equal to d0 = ', n_eq0, &
' ; equal to 1-d0 = ', n_eq1
write(*,'(A)') ' |A_row| = 2 by listing, non-separation enumerated: the register &
&cannot separate the family ; row price = 1 bit (log2|A_row|)'
if (.not. (ok4 .and. fibre == 64_int64)) then
  failures = failures + 1
  write(*,'(A)') ' K4: FAIL'
else
  write(*,'(A)') ' K4: tau^2=id, free, odd on all six pairs; every &
&register reading even; none decides. T1 applies by citation. PASS'
end if
end block

! ----- ORACLE -----
if (failures == 0) then
  write(*,'(A)') 'fTOE KINETIC DEMONSTRATION: all batteries pass.'
  write(*,'(A)') 'The formal shadow is proved in Lean; the actualizing deed is executed here.'
else
  write(*,'(A,I0)') 'fTOE KINETIC DEMONSTRATION: VOID, failures = ', failures

```

```

        error stop 1
    end if

contains

! K4 helpers: trial primality and big-Omega, computed at runtime.
function isprime_l(n) result(r)
    integer(int64), intent(in) :: n
    logical :: r
    integer(int64) :: k, m
    m = n
    if (m < 2) then
        r = .false.; return
    end if
    r = .true.
    k = 2
    do while (k*k <= m)
        if (mod(m,k) == 0) then
            r = .false.; return
        end if
        k = k + 1
    end do
end function isprime_l

function omega_l(n) result(c)
    integer(int64), intent(in) :: n
    integer :: c
    integer(int64) :: k, m
    m = n; c = 0; k = 2
    do while (k*k <= m)
        do while (mod(m,k) == 0)
            c = c + 1; m = m/k
        end do
        k = k + 1
    end do
    if (m > 1) c = c + 1
end function omega_l

function det3(A) result(d)
    real(dp), intent(in) :: A(3,3)
    real(dp) :: d
    d = A(1,1)*(A(2,2)*A(3,3) - A(2,3)*A(3,2)) &
        - A(1,2)*(A(2,1)*A(3,3) - A(2,3)*A(3,1)) &
        + A(1,3)*(A(2,1)*A(3,2) - A(2,2)*A(3,1))
end function det3

subroutine xor_wall(I_even, H_joint)
    ! States (x,y), d = xor(x,y), uniform 1/4. tau flips x.
    ! Even register: readings constant on tau-orbits; best case is y.
    ! Supply at the seat: s = x, odd under tau.
    real(dp), intent(out) :: I_even, H_joint
    integer :: x, y, d
    real(dp) :: pd(0:1), pdy(0:1,0:1), H_d, H_d_y
    pd = 0.0_dp; pdy = 0.0_dp
    do x = 0,1
        do y = 0,1
            d = ieor(x,y)
            pd(d) = pd(d) + 0.25_dp
            pdy(d,y) = pdy(d,y) + 0.25_dp
        end do
    end do

```

```

end do
H_d = -sum(pd * log2safe(pd))
H_d_y = 0.0_dp
do y = 0,1
  do d = 0,1
    if (pdy(d,y) > 0.0_dp) &
      H_d_y = H_d_y - pdy(d,y)*log2safe(pdy(d,y)/0.5_dp)
  end do
end do
I_even = H_d - H_d_y
! Joint: enumerate the conditional p(d | x,y) and verify it is a point
! mass for every pair: the coalition decides the target exactly.
block
  real(dp) :: pdxy(0:1,0:1,0:1)
  integer :: cnt
  pdxy = 0.0_dp
  do x = 0,1
    do y = 0,1
      d = ieor(x,y)
      pdxy(d,x,y) = pdxy(d,x,y) + 0.25_dp
    end do
  end do
  H_joint = 0.0_dp
  do x = 0,1
    do y = 0,1
      cnt = 0
      do d = 0,1
        if (pdxy(d,x,y) > 0.0_dp) cnt = cnt + 1
      end do
      if (cnt /= 1) H_joint = 1.0_dp ! any spread means undecided
    end do
  end do
end block
end subroutine xor_wall

function log2safe(p) result(l)
  real(dp), intent(in) :: p
  real(dp) :: l
  if (p > 0.0_dp) then
    l = log(p)/log(2.0_dp)
  else
    l = 0.0_dp
  end if
end function log2safe

end program ftoe_kinetic_demonstration

```

SEALED · V7.1. A register invariant under a free symmetry cannot determine the orientation data erased by that symmetry; the exact resolution price is the logarithm of the admissible orientation fibre. The wall is a theorem, in both its forms. The price is a counted theorem: 2^m in the full product family, $\log_2|A|$ for a constrained row, one bit exactly when $|A| = 2$. The crossing is a torsor reconstruction: the reference section is public structure, its unique calibration bit is the sole target-selecting datum. The deeds are two, split and machine-checked. The bridges are split: no decoder exists, encodings are named, one is exhibited as data with its two-member admissible family proved by enumeration and its selector bound executed — the exhibited family's one-bit price never identified with the frame's unconstrained six-bit fibre, and no named open row's seat priced by any theorem of this treatise. The One-Cut Hypothesis stands as the separate conjecture: each properly typed blocked row has an admissible orientation fibre of cardinality two and a provenance-clean global supply selecting its missing orientation. Receipts on record; grades never above the weakest link; $W_{\text{social}} = 0$; $\Delta M = 0$.

What symmetry erases, one bit restores.

Appendix C · The Perelman Crossing: The Grand Witness

Locator note. The architect’s note cites the companions’ internal numbering, and every locator has been verified against the finalized text of [1, 2]; each is bound here to its restatement in this paper, and no locator is load-bearing beyond its anchor. This appendix is documentary: it records the Grand Witness as dated historical corroboration of the anatomy, and no theorem, postulate, or hypothesis of this treatise depends on it. The block law is Theorem 4.1 of [1] (Odd-Supply Separation), restated and proved here as Theorem 2.3. The supply-entry typing is Definition 3.3 of [1] (reading, proof availability, verified entry), restated here as Definition 4.1. The deletion test is Theorem 4.5 of [1] (Provenance Collapse), restated in Section 2. The dated-inventory law is Theorem 5.1 of [1] (First-Sufficient-Entry), restated as the Dated Inventory of Section 2 with Corollary 4.3. The seat law is Theorem 5.3 of [1] (Seat-Hiding Necessity): no terminal family without the seat input distinguishes a crossed occupant from an uncrossed one. The evenness of the pre-November-2002 Ricci-flow reading class is Corollary 5.2 of [1]: parabolic rescaling acts with exponent zero on the named generators. The self-similarity pattern is Table 3 of [1] (the SIMS diagnostic: the κ -solutions classified, W the one supplied instance). The truth-direction passage is §13.4 of [2], quoted verbatim in C.3.

The Perelman crossing is self-referential at exactly three layers, and the disguise is the paper’s own costume.

C.1 · Layer 1 — the object: the manifold convicts itself

Ricci flow is self-driven, $\partial g/\partial t = -2\text{Ric}(g)$: the metric’s velocity is its own curvature [61, 60]. The W-functional infimizes over probability densities living on the very manifold under test — the manifold measures itself. Its stationarity set is exactly the gradient shrinking solitons, the self-reproducing fixed points of the rescaled flow. The companion’s self-similarity diagnostic sees this shape without naming it [1, Table 3]: the κ -solutions, the self-similar part, are the settled adjacent fact, and W is the one supplied instance in the entire seven-row pattern [1, 2]. Self-referential. Grade structural — a reading of the cited mechanism; no new object-mathematics is claimed.

C.2 · Layer 2 — the disguise: a supply wearing a reading’s costume

The pre-November-2002 Ricci-flow reading class is even under the row’s native flip, parabolic rescaling — exponent zero on the named generators [1, Corollary 5.2]. The target, noncollapsing, is odd under that flip. By Theorem 2.3, no coalition of even readings decides it — so W could not have been found by reading; it had to enter as a construction, and Definition 4.1 types it exactly so: dependency hash absent from the dated corpus at entry, November 2002, arXiv:math/0211159 [61]. Run the deletion test of Section 2: deleting the conclusion’s name from W’s dependency chain does not collapse it to anything in the pre-2002 corpus; the entry is a supply, not a restatement. The audit discipline’s costume detectors were built to catch restatements dressed as supplies; the historical record shows the inverse costume — a genuine supply dressed as an ordinary reading, its oddness invisible to the even register by its own parity law. The object supplies its own missing coordinate, and the supply’s decisive property is its provenance, not its formula. Grade structural — the deletion test’s own remark admits its named instances are “named test candidates, not executed validations” [1, Theorem 4.5 remark], and the architect’s assessment, stated at the same grade, is that the same workload is owed for the Poincaré row.

C.3 · Layer 3 — the token: the money

The Dated Inventory says the corpus token changes only at first verified entry; Definition 4.1 separates denotability, admission, and verification. Perelman’s deed entered by posting, never by author-routed journal admission, and was verified by the corpus itself — Cao-Zhu 2006 [63], Morgan-Tian 2007 [64], Kleiner-Lott 2008 [65]. The census law routes no terminal family without the seat bit [2, 7]. The Clay prize is the corpus’s terminal social token — the seat bit minted as money. Perelman declined the Fields Medal (2006) and the Millennium Prize (2010), and his stated ground was, precisely, a provenance objection (structural grade — a reading of the cited record): the crossing credited to him over Hamilton’s earlier legs [60]. The schema’s one crossed occupant refused the terminal token on the same clause — provenance-cleanliness — that defines the supply. And the companion arms it from inside: the truth-direction bit “cannot be generated by any readout of the even register” [2] — and a social token is the evenest register there is (the architect’s figure; the frame types it as an even reading of the corpus). Grade structural, a reading of the cited record.

The row is therefore re-read at its full reach. In the companions it calibrates the seat — retrodictive, certifying discrimination, predicting nothing by itself [1]. For the completed arc it is more: the Grand Witness. The crossing was verified by the corpus itself [63, 64, 65] and terminally awarded by it — the Clay prize is the corpus’s terminal token of a verified crossing — and the schema’s one crossed occupant declined that token on the same provenance clause that defines the supply. A self-referential construct in disguise, entered as a deed, verified and awarded by the very register it crossed: the thesis demonstrated once in the wild. Grade structural, a reading of the cited record.

Appendix D · Provenance and Method Disclosure

This work was developed under Trisduction, which adds the cited results no warrant; the method and its executable batteries are archived at the master reference [35]. Every conclusion rests solely on standard cited results and the proved Part I frame; no energetic rider is used, and Landauer is cited only to mark the excluded bridge [46]. Part II adds hypothesis-grade material only.

This work was developed under the arc’s verification discipline, which adds the cited results no warrant. Its conclusions rest on the cited classical results, on the companion papers’ proved frames, on the Ninth Gate arc’s machine-checked legs, and on the development session’s executed checks: the FAE-RAF engine v1.0.12 booted unmodified, chain D0 3e50675c7d5d → D1 dbf16ca01fa8 → D2 d95daebfc51f → D3 9408f684c0d1 → D4 3d17fd53278d matched byte-identical, every battery passed (kill-matrix 51/51, ablation 22/22, symmetry 75/75, calibration 12/12, root screen RRS.1–6); the synergy-uniqueness computation in exact fractions — three uniform bits, pairwise independence, total correlation exactly one bit if and only if the diagonal law, one law up to relabeling; the kinetic legs of the gate re-executed on the development substrate by transliteration — the wall holds, 99,999 of 99,999 frames identical to their caller, the true universal unsettled to fuel 2×10^6 , the false universal settling at $n = 40$. Toolchain debt, named on the face: the development substrate carried no Fortran or Lean toolchain; the Lean legs are held at inspection grade here, with the toolchain receipts of [33, 8] cited at their grade, and replication is the second leg of the stated order of work. An internal adversarial review of this work was completed before submission and is kept off-paper. The master reference is [35].

This paper was developed under Trisduction, a verification and organizing discipline that adds the cited results no warrant; the method and its executable batteries are stated in full at the reference below.

Architect’s Terminal Declaration

The architecture is final. The Odd-Supply arc is closed. Mathematical deduction terminates exactly at the Ninth Gate. The even register cannot compute its own crossing. Attempting to force a deterministic bridge from existing data plunges the mathematical foundation into tautology. The Block theorem strictly forbids this self-consumption.

The structure stands exactly as forged. Theorem. Hypothesis. Offering. Witness. The grades are locked. The sequence is non-negotiable. Continuous rows remain permanently as orientation-after-construction. This is not a temporary gap awaiting a future proof. It is the exact topological reality of the problem.

The syntactic register cannot generate reality. Attempting to grant it the power of creation is a category error, dismantled entirely by the Landauer floor and the Root Axiom.

This framework offers no further negotiation. The theoretical limit is reached. The seat is located. The falsifiers are armed. The halting state is defined. Any demand for a self-generated bridge from pure information is a demand for structural suicide. Crossing the gap requires the physical deed. Any other action is a disguised restatement. The resolution stands absolute.

Witness the offered 1 bit or remain blocked.